Instructions in preparing an application

Applicants are required to use this template to submit their Flagship program. Minimum font 11pts (Arial, Calibri, Times New Roman), standard line and character spacing, and normal margins. All indicated chapters are mandatory unless otherwise indicated. We recommend you respect the indications given in each chapter in order to adhere to the expected content. A word count is clearly indicated for each section. If you do not adhere to the maximum amount of words, your application will not be processed. The word count includes tables, footnotes and figure legends but excludes references. Please do not include hyperlinks. The application form needs to be submitted in a single pdf document, including the budget table and letters of intent.
We recommend that you do not send in your submission at the last minute in order to avoid technical difficulties. No submission will be accepted after the deadline, March 15th at 13:00 CET.
Applications should be sent in PDF format to ***Flagships@erasmusmc.nl***.

Title: ***Rare Disease Innovation Center (RD-IC) initiative: Converging stakeholders and sciences to propel the health journey***

Rare Disease
Innovation Center

Health & Technology
By Convergence

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

*Name**	*Institution*	*Department*	*Email*	*Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)*	*For Erasmus MC applicants: clinician/ researcher/ mix*	*Role in Flagship [Main Lead/Lead/ Applicant/Co-applicant]*
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Dr. A. Albayrak (WP2 TUD lead)	TU Delft (TUD)	Industrial Design Engineering	a.albayrak@tudelft.nl	Tenured	N.a.	Lead
Dr. T.S. Barakat (WP3 EMC lead)	Erasmus MC	Clinical Genetics	t.barakat@erasmusmc.nl	Tenured	Researcher/ Clinician	Lead
Prof. dr. C.T.B. Ahaus (WP2 EUR lead)	EUR	ESHPM/ Health Services Management & Organization	ahaus@eshpm.eur.nl	Tenured	N.a.	Co-lead
Dr. J.C. van Gemert (WP3 TUD lead)	TU Delft	Intelligent Systems	j.c.vangemert@tudelft.nl	Tenured	N.a.	Co-lead
Dr. T.J. van Ham (WP3 EMC lead)	Erasmus MC	Clinical Genetics	t.vanham@erasmusmc.nl	Tenured	Researcher	Co-lead
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		Policy & Management				
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Patientenfederatie Nederland	*Approached and interested*	N.a.	m.degouw@patientenfederatie.nl	N.a.	N.a.	Advisory Board
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NWO-ZonMW	*Approached and interested*					Advisory board
Pharos	*Approached and interested*	Health literacy	g.boland@pharos.nl			Advisory board
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Prof. dr..H. Kayserili Karabey	Koç University School of Medicine (KUSoM) Koç University Hospital, Istanbul	Professor of medical genetics, rare disease expert	hkayserili@kuh.ku.edu.tr	N.a.	N.a.	Advisory Board
Kees van den Berg	GenomeScan BV	CEO and Founder	k.vandenberg@genomescan.nl	N.a.	N.a.	Advisory Board

- Corresponding Lead/ Main applicant: Dr. M.H. Cnossen. Other leads: Dr. T.S. Barakat, Dr. A. Albayrak; Co-leads: Dr. T. van Ham, Dr. R. Schappin, Prof. dr. C. Uyl-Groot, Dr. K. Ahaus, Dr. J. Rezaei, Dr. J. van Gemert.
- WPs consist of EMC-TUD-EUR partners, one institute is leading per WP and underlined in applicant list.
- Planned start date of seed funding: *1-9-2022*
- Primary Convergence theme: *Improving Health Journeys*
- Secondary Convergence theme (*if applicable*): *Technology Supported Transitions in Health(care)*
- Tertiary Convergence theme (*if applicable*): *Human Centered Technology and AI for Health*
- Quaternary Convergence theme (*if applicable*): *Fundamentals of Health & Disease*

Keywords: *please list a maximum of 7 keywords*
1) Rare diseases
2) Health journey
3) Life course medicine
4) Genetics
5) Patient journey
6) Decision support by multi-attribute analysis
7) Precision medicine

1.b Summary (298 words)

Improving health(care) in rare diseases (RDs) is urgent and a major societal challenge as RDs together affect 6-8% of Europeans. In the Netherlands, >1 million people have RDs. Many RDs reduce lifespan, diminish quality of life and societal participation due to lack of: a) accurate, timely diagnosis; b) informed decision-making support ensuring optimal diagnostics and treatments; c) effective, affordable therapies; and d) societal awareness of the need for better support and societal participation for individuals-with-an-RD. Despite (inter)national initiatives, progress addressing these unmet needs is slow. Solutions are complex, requiring interdisciplinary collaboration, bundling resources and involving key stakeholders. Based on their unique combination of knowledge, research capacity and resources, Erasmus MC (EMC), EUR and TU Delft (TUD) are outstandingly positioned to establish such multidisciplinary collaboration, advancing RD-health journeys for individuals-with-RD and society.

The goal of the ***Rare Disease Innovation Center initiative*** is to improve the RD-health journey (**Figure-1**). We aim to:

- **WP1**: Connect and support RD-communities by creating an **RD-health data-platform** connecting all stakeholders (RD-individuals, patient-organizations, clinicians, RD-centers, scientists, policy makers, insurance companies).
- **WP2**: Improve **RD-individual care pathways** to become cost-effective and inclusive, with attention to mental health, health literacy and ethnic diversity, by applying data-driven research, experience-based co-design, integrating WP3/WP4/WP5 innovations.
- **WP3: Advance innovative RD-diagnostics** increasing diagnostic yield by implementing whole genome sequencing, RNA-seq, Med-seq and proteomics, integrated with artificial intelligence-based syndromic image-recognition.
- **WP4: Install diagnosis and treatment decision support** using multi-attribute analysis.
- **WP5: Facilitate implementation of innovative, affordable RD-therapies** (gene, cellular, compound), fostering entrepreneurship.

By its goals and design, **this Flagship will directly improve the health journey of RD-individuals, increase societal awareness, and advance interdisciplinary academic excellence in new research fields bridging basic-to-clinical research, branding the triad EMC-TUD-EUR as spearhead of (inter)national RD-research.**

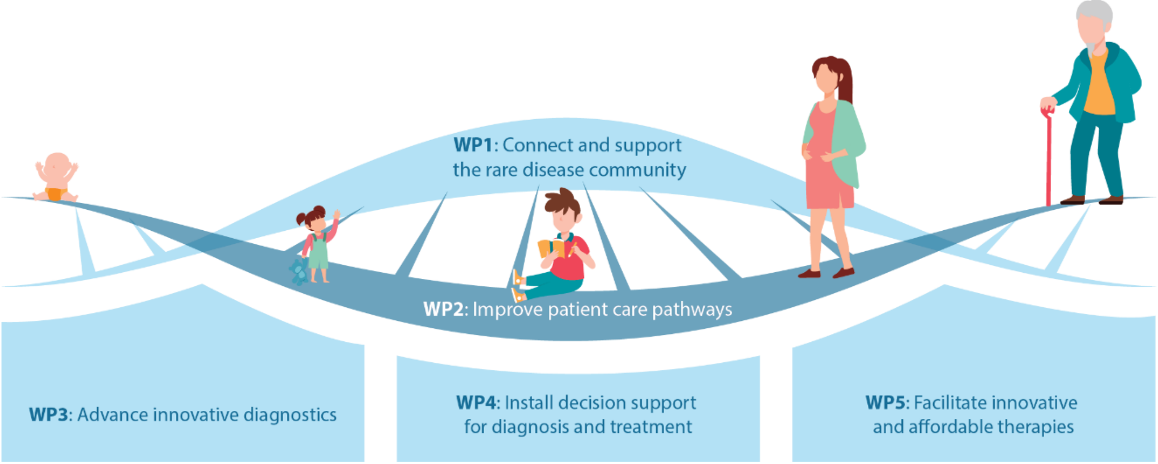

***Figure-1**. Improving the RD health journey by joining forces and enhancing implementation of innovations*

1.c Public Summary (300 words)

Meer dan 1 miljoen Nederlanders hebben een zeldzame ziekte. Deze mensen hebben gezondheids- en psychische problemen, leven vaak korter en kunnen niet goed meedoen in de maatschappij. Velen weten niet waar ze voor goede zorg moeten zijn. Het is daarom belangrijk om de zorg voor mensen met een zeldzame ziekte te verbeteren. Dit kan door beter vindbaar te zijn; nauwkeurigere en snellere diagnoses; duidelijkere keuzemogelijkheden voor patiënten en artsen; goed werkende, betaalbare behandelingen; en meer aandacht hiervoor in onze samenleving. Vanuit de politiek werkt men hier de laatste jaren hard aan. Toch duurt het te lang om dit beter in te richten. Het gaat om ziektes met ingewikkelde zorg. Echter, als mensen met een zeldzame ziekte, zorgverleners en wetenschappers uit verschillende vakgebieden samenwerken, boeken we meer vooruitgang. Een samenwerking tussen de zorgverleners en wetenschappers van Erasmus MC, TU Delft en de Erasmus Universiteit verbindt de kennis, ervaring en de middelen om écht iets te kunnen betekenen. Daarom willen wij in dit project met elkaar aan de slag.

We noemen onze samenwerking het ***Rare Disease Innovation Center***, en gaan:

- Alle mensen en organisaties die betrokken zijn bij zeldzame ziekten samenbrengen en een digitale plek maken waarop ze elkaar kunnen vinden en ontmoeten.
- Ervaringen van patiënten verbeteren. Iedereen moet gebruik kunnen maken van speciale zorg en uitvindingen die de zorg verbeteren, moeten zo snel mogelijk worden toegepast.
- De diagnose van zeldzame ziekten nauwkeuriger en sneller maken door nieuwe laboratoriumtechnieken en kunstmatige intelligentie.
- Patiënten en artsen helpen beter te kiezen door het gebruik van wiskundige modellen.
- Helpen bij onderzoek naar nieuwe behandelingen en hoe deze betaalbaar kunnen blijven.

Met deze doelen en onze samenwerking willen wij één van de belangrijkste organisaties voor onderzoek naar zeldzame ziekten in Nederland en Europa worden, zodat we de toekomst van mensen met een zeldzame ziekte kunnen verbeteren.

2. Relevance to Convergence Health & Technology mission (50%) – max. 2500 words for 2.a to 2.e

- The Flagship program contributes to the Convergence Health & Technology mission (1)
- The Flagship program objectives fit with the guidance per theme provided by the theme leads, and/or fit with more than one theme of Convergence Health & Technology (2)
- The Flagship program merges approaches and insights from distinct disciplines to reach its goals (3)

• The staff granted by this call actively brings together different disciplines within the program, and creates visibility and attracts talents, partners, and funding for the program (4)
• The Flagship program has a clear view on long term sustainability after the 5-year central funding period (5)
• The envisioned results of the Flagship program have the potential for a significant contribution to healthcare, society and/or have scientific impact which has the potential for long-term societal impact (6)
• The Flagship program provides opportunities for BSc/BA and/or MSc/MA students of the participating institutes to participate in (interdisciplinary) projects (7)
• The proposal defines the pathway of how this contribution can be realized. The pathway includes concrete research and/or impact pathway (societal and/or economical) valorization strategies (8)

2.a Contribution to H&T mission and fit with H&T theme(s) (454 of 2475 words)

Our ***Rare Disease Innovation Center (RD-IC)*** Flagship proposal aligns to H&T themes: *Improving Health Journeys,* but also contributes to *Technology Supported Transitions in Health(care)/Human Centered Technology* and *AI for Health*. RD-IC will directly improve the health journey of RD-individuals, increase societal awareness, and advance interdisciplinary academic excellence in new research fields, with an interdisciplinary mindset, bridging basic-to-clinical research while educating younger generations.

Promoting life-long health and well-being of RD-individuals
We will ***innovate the RD-health journey by integrating experience-based co-design with knowledge and viewpoints of crucial stakeholders*** (e.g., EMC-led Health-RI, PHARMO Institute), and public. Our ***life-course approach*** will consider the needs of RD-individuals and their families, and RD-specific health events such as genetic counseling, (lack of) diagnosis, treatment initiation, transitions in care, and life-long care.

RD-individuals are treated in 57 Dutch Federation of University Medical Centers (NFU) RD-centers in EMC and ***(inter)national RD-centers/European Reference Networks (ERNs)***. By applying ***fast-track innovation***, we aim to create long-term impact on health and well-being of all RD-individuals worldwide, and a) improve the diagnostic yield, leading to enhanced ***personalized health(care) and precision medicine opportunities***, targeting disease-specific biological pathways[1-7]; b) apply decision-making support using multi-attribute analysis addressing costs/benefits of diagnostic and therapeutic choices; c) implement clinical application of innovative, affordable therapies; which d) when combined with psychosocial interventions will improve RD-individual/family wellbeing and societal participation, monitored by patient-reported outcomes[8]. These innovations and expertise will be implemented in our teaching at EMC-TUD-EUR (**section-2c**).

Promoting socio-economic equality in RD-health(care)
We will ***reduce RD-health inequalities*** by improving access to specialized RD-healthcare for families with ***limited health literacy skills and/or diverse cultural backgrounds***[9-13]. We will investigate what underlies these inequalities in RD-care and develop approaches how to address them (**WP1**/**WP2**). Furthermore, we aim to make Caucasian-based genetic reference datasets more ethnically diverse (UNITED initiative/Adams EMC), reducing bias in test results for individuals of other descent (**WP3**). Furthermore, socio-economic inequalities are pivotal in *MSc Genomics in Society* (**section-2c**).

Improving prevention by early RD-diagnostics and individual-tailored medical treatment

For many RDs, an ***early (genetic) diagnosis*** accelerates effective medical interventions, ***preventing illness or mortality***, e.g. specific diets for inborn errors of metabolism[14]. We aim to implement diagnostic innovations including ***functional genomics***[15-22] (**WP3**) and use multi-attribute analysis[23-27] to develop decision-support for personalized diagnostic and treatment pathways (**WP4**). Ongoing central data collection will allow for continuous fine-tuning and access to population health-data will aid development of prognostic models (**WP1/WP2**). Designing a technology-driven RD-care ***precision medicine clinic***, with rapid implementation of diagnostic and treatment innovations such as gene therapy (lenti-/adeno-associated viral), CRISPR/Cas9-based therapies, and other biologicals (**WP5**), will directly promote RD-individual health. Our design will serve as a blueprint for other RD-centers in line with Convergence pillar fast-track innovation. We will work in alliance with other related Flagships such as *iMed/Neuromedicine/BioComm.*

2.b Synergy between disciplines (514 of 2475 words)

*Elaborate on how this program **merges approaches and insights from distinct disciplines** to reach its goals.*

The challenges to improve the RD-health journey are too complex to be tackled by a single research discipline or institute. This complexity is due to multi-faceted interdisciplinary challenges. Convergence H&T has a unique advantage compared to prior RD-initiatives: ***synergy between world-class technology, healthcare economics, social, and legal science-centered universities, and an excellent academic hospital***. Consequently, the RD-field will particularly benefit from the ***interdisciplinary approach and mindset*** of our Flagship *with expertise from >20 departments in this Flagship:* e.g. RD clinical expertise (Pasmans, Cnossen/EMC), will be combined with social sciences (Ahaus, Schappin/EUR+EMC), health care economics (Uyl/EUR), genomic and diagnostics (van Ham, Barakat/EMC), computational sciences (Reinders, van Gemert/TUD), decision making (Rezaei/TUD) and clinical therapy development (Pijnappel/EMC) together yielding the unique multi-faceted expertise required to achieve our aims and improve the RD-health journey (**Figure 2, section-4**).

Synergy within our Flagship is further enhanced by overlapping postdoc and PhD-positions between WPs and disciplines (**section-2c, Figure-4/Table-1**) expanding knowledge dissemination. This process, amplified by the geographical vicinity of EMC-TUD-EUR, will create scientific impact for all involved, especially RD-individuals, but also trainees (postdocs, PhDs, BSc/BA, MSc/MA-students) (**Figure-3**).

Importantly, our Flagship is an ***interesting partner*** for other ***external public and private entities*** which also aim to accelerate innovation, e.g. Data Analysis and Real-World Interrogation Network (DARWIN-EU, led by EMC) coordinating medication studies for European Medicines Agency (EMA), and Future Affordable and Sustainable Therapies (FAST), a national platform investing in future therapies and infrastructure. RD-IC is the ideal hub for clinical trials in which RD-populations are accessible and well organized.

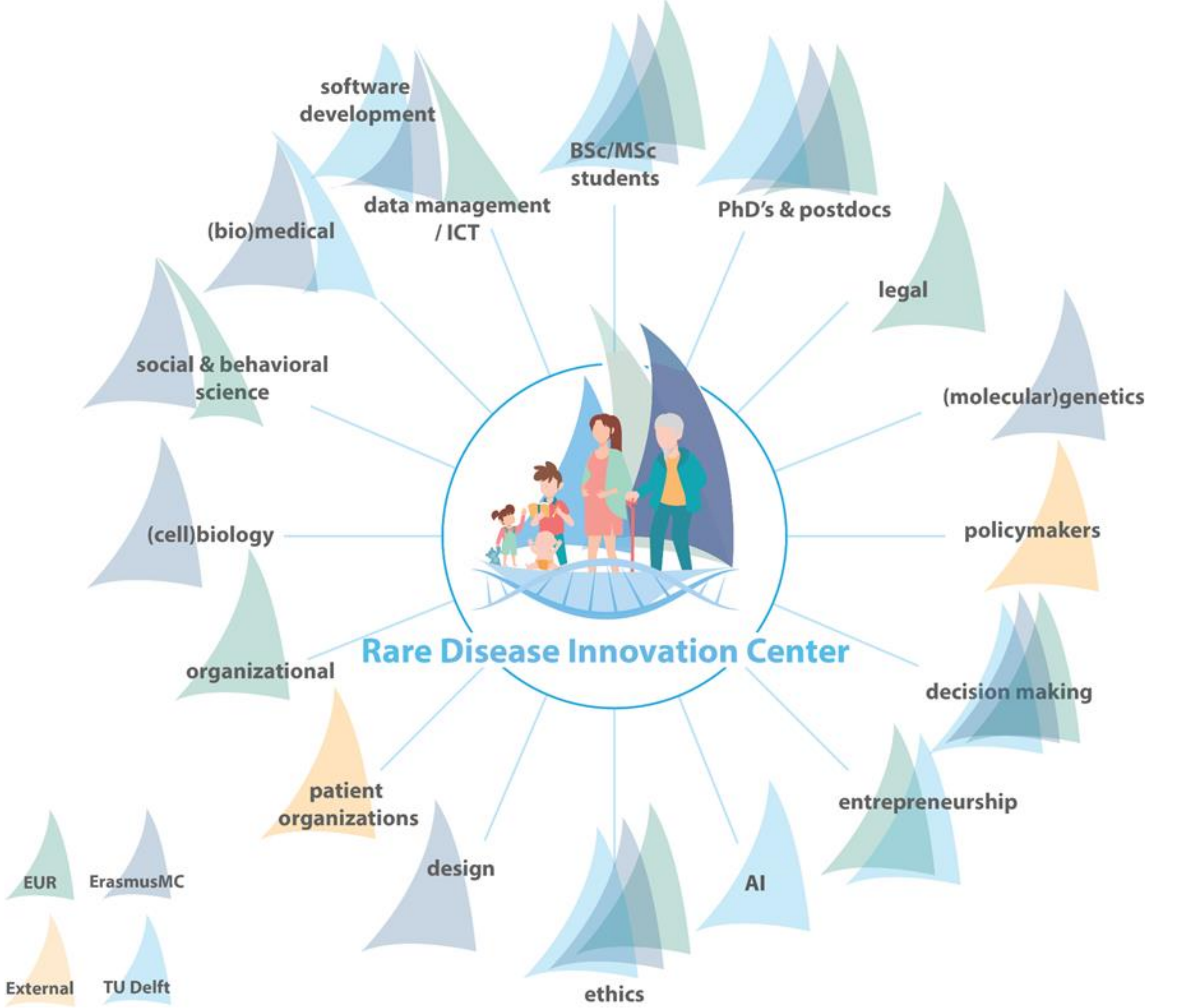

***Figure-2**. Flagship as a multidisciplinary, multi-stakeholder entity improving the RD-health journey, empowering RD-individuals and propelling RD-research.*

We believe our Flagship can create societal impact for RD-individuals and society at large as **EMC i**s already an important player in the care for RD-individuals. It headquarters an ***impressive 57 of >300 NFU RD-centers in the Netherlands***, the most of any Dutch academic hospital. These RD-centers have been initiated by the Ministry of Health to align with the *European Union's (EU)* and *European Organization for Rare Diseases' (EURORDIS)* declaration to identify RD-expertise and advance research[28-29]. Certification criteria for RD-centers align with our Flagship, and include multidisciplinary (para)medical teams, treatment of both children and adults using protocolled care pathways, and (translational) research[30]. EMC also ***participates in 23/24 European Reference Networks (ERNs)***, leading two of them (**Figure-3**). Locally, EMC RD-centers aim to unite, to enhance knowledge dissemination by collaboration and synergy. A promising beginning has been made with the EMC-Sophia Children's hospital's focus area for RDs, chaired by van der Ploeg. RD-IC can further propel this alliance.

Such an alliance is badly needed due to the challenges we are confronted with despite European initiatives. Challenges include: Fragmentation of expertise and resources leading to poor access of specialized care, delayed diagnoses and lack of effective therapies;

research groups working with limited resources, in small patient cohorts; and unaligned research due to lack of overarching funding. Drastically, in 2021, a UN resolution (A/RES/76/132) addressed these challenges and human rights of RD-individuals/families, calling for worldwide action. Together, EMC-TUD-EUR will synergistically impact the RD-field overcoming these challenges.

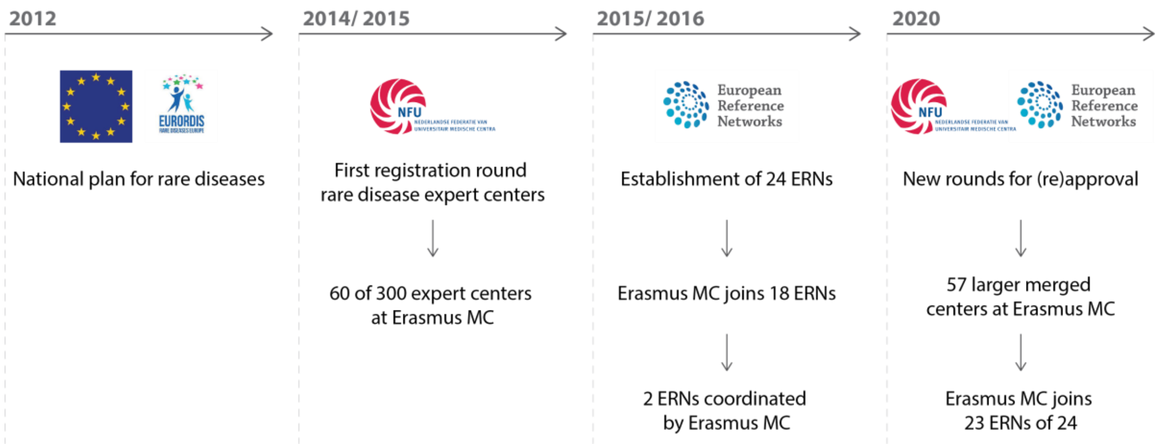

Figure-3. *European and national RD-initiatives*

2.c Long term sustainability plan (773 of 2475 words)

*Please indicate your view on the feasibility of the program as well as the **long-term sustainability** plan of the program **after the 5-year central funding period**. Describe **possible risks** and how to tackle them.*

Our Flagship program will support a ***growing and sustainable Rare Disease Innovation Center (RD-IC),*** with long-lasting impact on the RD-field, feeding international research agendas by its long-term innovative research-strategy. Dedicated RD-research will remain required for decades to come as there are ~8000 RDs, together affecting ~300 million individuals/families worldwide, many of whom currently experience severe (mental) health problems due to limited health(care) options and estimated to cost €698 billion of health care costs annually in the US only (2016)[31]. Personal and steady public and private partnerships will retain sustainability over the years. Importantly, the impact of RD-discoveries, extends beyond RDs alone, impacting also common diseases, by uncovering general pathophysiological mechanisms, which can be modified or intervened.

Research and design of the consortium

We present a feasible and detailed five-year program with deliverables (**section-3c**; **Figure-6**). ***Sustainability will be supported by knowledge dissemination and structured interactions between WPs***, via shared postdocs/PhD-positions (**Figure-4/Table-1**) and cross-talk with educational programs. This unique setting will attract talent, from students to renowned (inter)national senior scientists, catapulting RD-knowledge and funding potential.

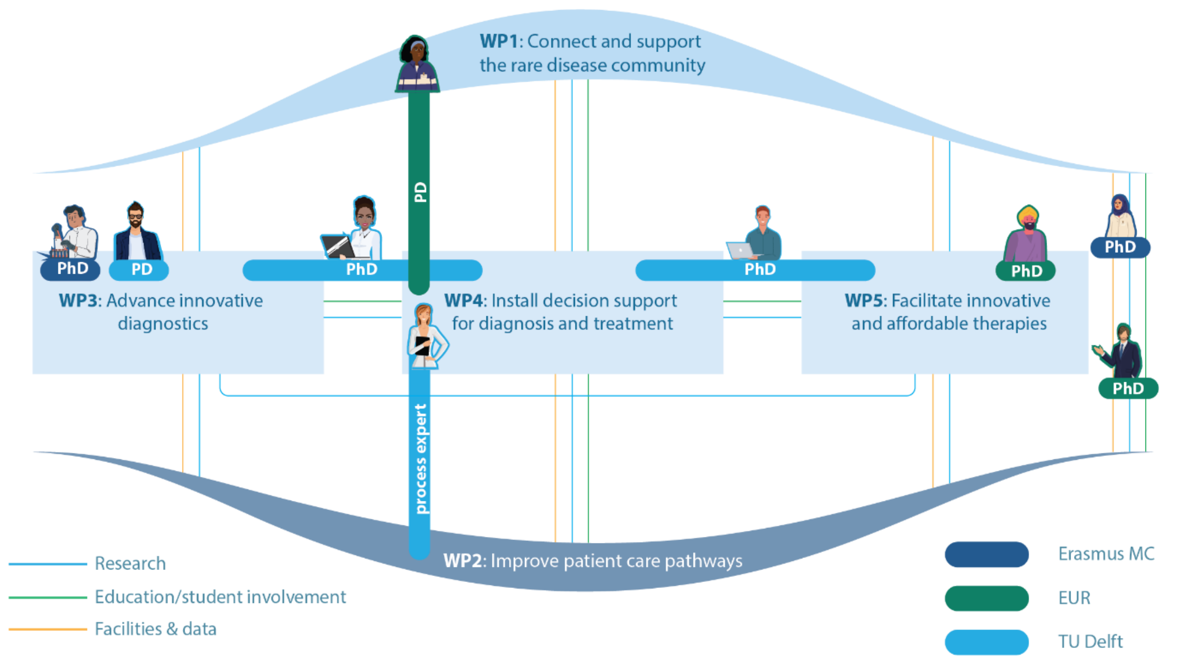

Figure-4. *Convergence and shared positions between WPs in the five-year Flagship program; also see Table-1. PD=postdoc, PhD=PhD-student.*

	WP1	WP2	WP3	WP4	WP5	Total (€)
<5 years +€2 million	0.4-fte postdoc (WP1/4;EUR)		0.8-fte postdoc (ICT;TUD)	0.4-fte postdoc (WP4/1;EUR)		2-fte postdoc/2yrs
	0.5-fte PhD (WP1/2;MD;EMC)	0.5-fte PhD (WP1/2;MD;EMC)	0.5-fte PhD (WP3/4; TUD;3 yrs)	0.5-fte PhD (WP4/3; TUD;3yrs)	1-fte PhD (Business; EUR)	6 PhDs/of which 3/3yrs
	0.5-fte PhD (WP1/2;HTA/EUR)	0.5-fte PhD (WP2/1;HTA/EUR)	1-fte PhD (EMC;3yrs)	0.5-fte PhD (WP4/5;3yrs)	0.5-fte PhD (WP5/4; TUD;3yrs)	

Table-1. *Overlapping postdoc/PhD-positions accelerating synergy and sustainability, supporting scientific and financial ambitions. Postdoc (2yrs); PhD (4yrs); MD=medical doctor; HTA=Health Technology Assessment*

Education and training of young researchers, by involving students directly in our research, is essential. RD-IC has strong ties with education in EMC-TUD-EUR: participating clinicians/scientists (>18 BKO/UTQ; university teaching qualification-certified) are actively involved in >5 MSc, 3 BSc-programs and ~50 BSc/MSc/PhD-courses encompassing all disciplines in RD-IC. We co-organize EUR MSc Health Economics, Policy & Law, including courses on HTA (treatment RDs with expensive medicine/reimbursement), EMC BSc/MSc (molecular)medicine programs and TUD Faculty Engineering, Mathematics and Computer Science (computer Vision, deep-learning, decision-making methods). The latest addition is

EMC Research MSc Genetics in Society (GiS, 4 founders in RD-IC). Engaging in our Flagship program is a unique chance for GiS-students/RD-IC researchers to work together on key RD-IC topics. We will also install a learning-environment focused on intellectual growth of young scientists/clinicians, involving students in our WPs to explore new scientific niches, and allowing us to foster the new generation of RD-IC research leaders.

Funding roadmap and talent policy

Funding is key to our sustainability, therefore with Research Development Offices (RDO) of EMC-TUD-EUR, we will pro-actively initiate a *Funding-Roadmap* of (inter)national grant opportunities based on (inter)national research agendas and will initiate personalized strategies for tenure track staff focusing on multi-year grant application plans. The Roadmap will a.o. include: foundations, EJP-RD, NWO-for large scale infrastructure, NWO-Zwaartekracht, NWO-NWA, Health Holland, Horizon Europe and a Marie Sklodowska-Curie COFUND action. We believe that applying on behalf of RD-IC raises funders' attention and increases awarding of grants. We will create €1.2 million matching for year 3-5 as required and grow towards a generated annual budget of >€2.0 million euros, leading to >€10 million in +10 years (>8 novel PhD positions/year), which is a solid financial base to fulfil our ambitious goals.

To ***attract and bind talent***, we will implement research-career paths and mentoring for talented core scientific staff and talented students/postdocs/PhDs and stimulate participation in grant writing/career-development courses. Applications for personal grants to secure RD-IC research lines will be supported by experienced grant-writers from within RD-IC (VENI, VIDI, VICI and ERC-grant laureates are Flagship-members).

Possible risks and mitigation strategies

Although we see many opportunities for (co-)funding, given the (inter)nationally recognized importance of RD-research and track-record of obtained financing by RD-IC members (**section-4b**), lack of funding remains a potential risk. We therefore have a strategy to minimize brain-drain and to create a talent binding/funding roadmap mentoring system to identify candidates for specific grants. Potential other risks involve the complexity of RD-healthcare, including RD-individual related problems (small numbers, >8000 different RDs with diverse symptoms and complications, varying inheritance); many stakeholders with different priorities, CE quality marking of novel instruments and tests. Specific focus should be on privacy rules to avoid the identification of individual RD-individuals based on the uniqueness of their RD; on organization and legislation of data collection and sharing; and an overall potential resistance to change and innovation. As a Flagship, the RD-IC will have a central position, facilitating our efforts to characterize the RD-landscape in detail and actively connect stakeholders (**WP1**) to identify these challenges and anticipate timely solutions.

2.d Visibility, attracting talents and new partners (332 of 2475 words)

*Describe how you will create **visibility for the flagship program**, how you will **engage students** (BSc/BA and/ or MSc/MA) and early career researchers in interdisciplinary projects, as well as how you plan to **attract talents**, partners and funding.*

High ***visibility*** of RD in general and the RD-IC in particular will promote societal awareness of the major challenges for RD-individuals and long-term RD-IC sustainability and funding potential. To maximize visibility, the overarching **WP1** will specifically address 1) a sustainable digital platform for RD-stakeholders; 2) connecting professionals and RD-individuals; 3) effectively disseminating knowledge and data; 4) involving and empowering RD-individuals and professionals using a personal health record in care and research.

We will further increase visibility by:

- Being an important ***partner*** of for example: (inter)national initiatives to organize healthcare and research data (*Health RI; PHARMO/INSZO; Health Outcomes Observatory; H2O*), to increase safety of medication trials (*EMA/DARWIN-EU*), and to promote development of sustainable and affordable therapies (*FAST*).
- Positioning of our ***members in (inter)national (scientific) boards*** and projects which will emphasize our scientific leverage.
- Regular ***press releases, communication*** with the public through platforms e.g., *Amazing Erasmus* and *Delft Integraal*, social media, and public debates with relevant stakeholders, in addition to scientific (open access) publications/presentations at (inter)national conferences.
- Initiating excellent ***education***. Directly integrating BSc/BA/MSc/MA-students into the Flagship (**section-2c**), and advertising our research within educational programs. RD-IC members are pivotal in various EMC-TUD-EUR research programs (**section-2c**).
- Emphasizing our impact and ***attracting talent***, through a) annual RD-IC Young Investigator Awards for research achievements at BSc/BA, MSc/MA and PhD-student level; b) applications for specific scholarships e.g., Marie Skłodowska-Curie COFUND action (with *BioComm* Flagship (**section-2c**)).
- Promoting ***funding activities*** (**section-2c)**. Also, exploring ***fundraising/crowdfunding possibilities*** for RDs, together with patient-organizations and public/private partners, to create awareness of RD-challenges.
- Organizing an annual 2–3-day ***RD-IC conference***: to optimize knowledge dissemination and funding. The first three years, we will present ongoing research on day-1, and dedicate day-2 to novel collaborations and funding opportunities. After three years, we will add a third (inter)national day, open to the public/RD-stakeholders with free admittance for students.

Together, these different actions will ensure broad visibility of RD-IC, maximize recruitment of (inter)national talent, expand our network increasing our funding potential and generating a future-proof sustainable learning and research environment.

2.e Long term societal impact (402 of 2475 words)

*Describe the potential and possible pathways for a significant contribution to healthcare, society and/or scientific impact which has the potential for **long-term societal impact**. Pathways include **concrete research and/or impact pathway (societal and/or economical) valorization strategies**. If applicable, indicate expected TRLs and necessary partners. In case of fundamental research please describe the scientific impact and time scale of expected societal impact.*

What output will be delivered?
Within five years (**Figure-6**, **section-3c**), we have defined spheres of influence and stakeholders. Research in **WP1** has built a vibrant research network and a gateway to maximize societal impact for all involved (**Figure-5**). Output of **WP2** consists of implemented RD-(mental) health outcome measures, RD-care pathways including psychosocial care designed according to RD-individual experiences, a model for cost-effective pharmacokinetic-guided dosing of drugs, a generic calculation of life-course care consumption with evaluation of current payment models. **WP3** increases diagnostic yield, may provide prognostic markers, and provides insights into detected DNA variants and their effects, combined with facial and Gestalt recognition. **WP4**'s deliverables are a complete stakeholder overview and a decision-support system for shared decision making in diagnostic and therapeutic care pathways. Finally, pricing and entrepreneurship models for RD-therapies will be the central outputs of **WP5**.

How will the Flagship change the RD-health journey?
Completion of tasks, reaching milestones and accomplishing deliverables as described in **section-3c** will directly impact the RD patient-experience and health journey by: (1) improving access and inclusion in RD-healthcare, (2) yielding faster and more efficient RD-diagnosis, (3) allowing more personalized RD-treatment, (4) leading to better (mental) health of RD-individuals, (5) increased societal participation, and (6) considerably less undiagnosed RD-individuals.

TRL-levels: WP1, a generic personal health record, TRL3>7-8; WP2, a tool to design care pathways/patient-journeys TRL3>7-8; Some of the methods in WP3/4 are at TRL4-6, which we will further develop to TRL7-8, possibly 9. Integrating datasets for improved yield and decision making is currently at TRL1-3, and we aim to develop it into TRL7-8. WP5 focuses on identifying critical/optimal steps for reaching TRL9.

What is the societal, economic and scientific impact?
We will improve the RD-health journey, increase societal awareness, and advance interdisciplinary academic excellence in new research fields, branding the triad EMC-TUD-EUR as spearhead of (inter)national RD-research. We gain insight into complex healthcare costs of the RD -health journey and concomitant implementation strategies. Unnecessary and ineffective diagnostics are terminated (de-implementation) and aligned; purchasing and reimbursement strategies concerning expensive medicine result in better pricing and return on investments. Long-term impact includes increasing public support for cutting-edge research on innovative diagnostics and therapies, smart decision making, psychosocial interventions, and intelligent use of expensive orphan drugs. Impact can be seen as valorization of RD-care pathways. Revenue from licensing of new tests, procedures or therapies will be re-invested in research. To protect intellectual property, RD-IC will liaise with our TTOs.

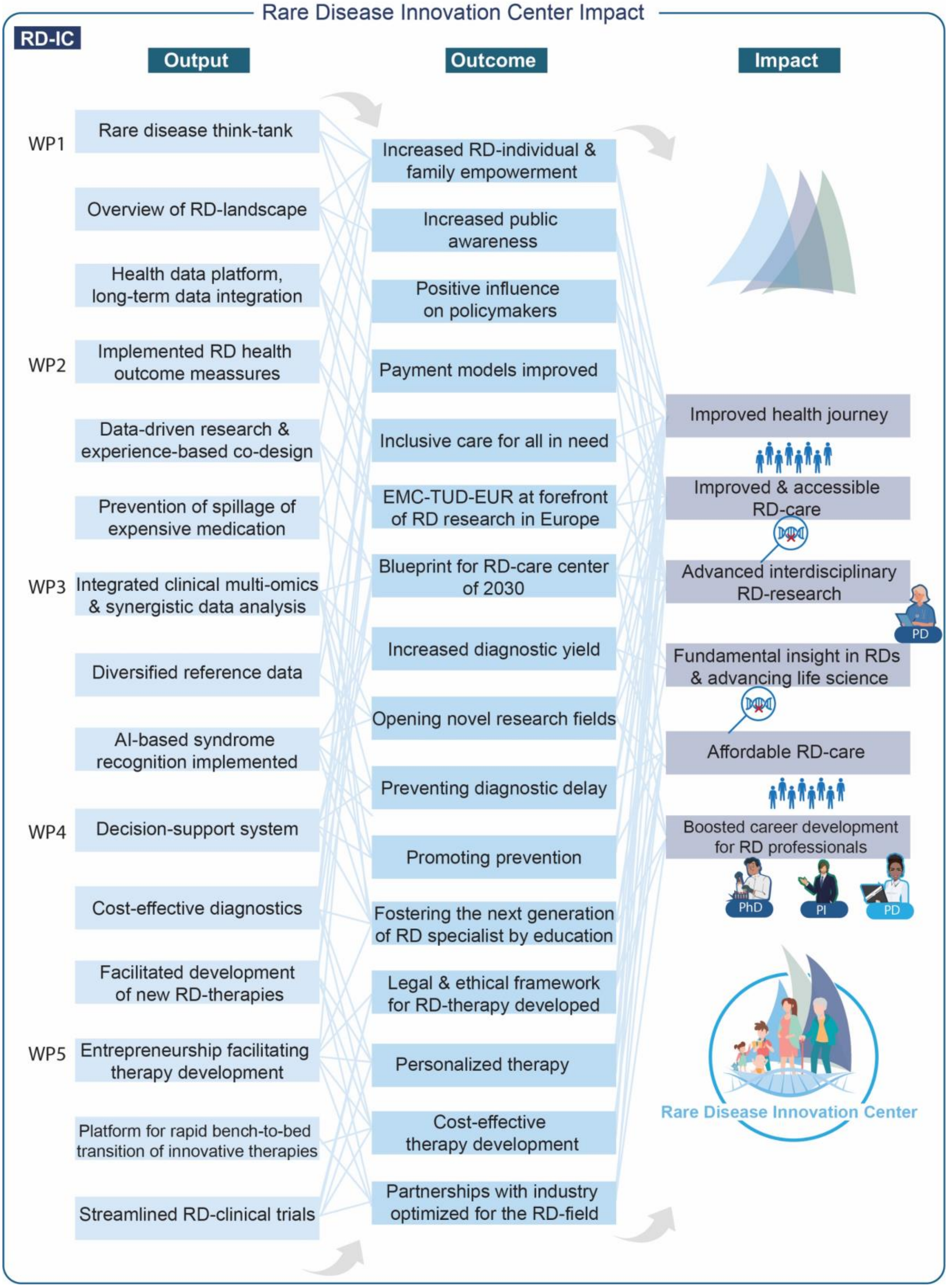

***Figure-5**. RD-IC Impact*

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3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

- The Flagship program contributes to excellent Convergence Health & Technology research, at the forefront of international research (1)
- The Flagship program is both innovative as well as challenging in its research and approach (2)
- The Flagship program has a clear strategy to engage students in its research plan (3)
- The Flagship program has a clear aim and long-term work plan, including, if applicable, a specification of required facilities and a data-and infrastructure plan. Coherence within the program is clearly demonstrated (4)
- The Flagship program aligns with or builds on (inter)national research agendas in this field, fit with the (inter)national research agendas but has a "high risk, high gain" main objective that is thoroughly explained (5)
- The Flagship program is feasible in terms of research performed, staff and budget needed (6)
- The Flagship program has a concrete plan for attracting external funding, which should result in at least the same amount as the central funding over the 5-year period (7)

3.a Ambitions (164 of 1998 words)

*Describe how the **ambitions** of the flagship program are in line with Convergence Health & Tech research. Please elaborate on the **innovative and challenging** aspects of the research program and/or its approach.*

Our **ambitions** are to lay a strong RD foundation at EMC-TUD-EUR and strengthen (inter)national and local initiatives to advance RD-care. RD-IC will advance interdisciplinary academic excellence in new research fields.

Our aspired outcomes are: (1) an interconnected RD-landscape, with methodological and data support for researchers and personal health records for RD-individuals, with access and dissemination of RD-care using (2) effective patient-centered health journeys that are supported by RD-individuals and clinicians with continuous outcome monitoring of (mental) health, (3) multi-omics and AI supported Gestalt recognition that diagnose >75% of RD-individuals, (4) decision support for RD-individuals and clinicians enabling personalized medicine whilst weighing societal interests such as cost-effectiveness, and (5) personalized and accessible genetic RD therapies. These highly innovative and interdisciplinary aspirations will greatly improve RD-individuals' (mental) health and longevity, as well as inspire young researchers. It will set the EMC-TUD-EUR triad at the forefront of RD-innovations in Europe.

Our aims and contributions to Convergence H&T are described in **section-2a**, our long-term societal impact in **section-2e**.

3.b Aims and feasibility (266 of 1998 words)

*Describe the **program's aims**, including a clear **strategy to involve students in its research plan**. Elaborate on the program's **feasibility** in terms of research performed, staff and budget needed. Address a concrete plan for **attracting external funding**. Also address **possible risks** and how to tackle these.*

Specific aims

1) Connect and support the RD-community with a health data platform connecting RD-individuals and advocacy groups, healthcare professionals, NFU RD-centers, scientists, data analysts, and policymakers to ensure connectivity and to explore the current RD-landscape and areas for improvement (**WP1**).
2) Improve RD care pathways after characterization of current fragmented RD patient-journeys, with attention for mental health, lower health literacy and ethnic diversity, by applying data-driven research and experience-based co-design (**WP2**), incorporating innovations from WP3/WP4/WP5.
3) Advance innovative RD-diagnostics, improving the diagnostic yield by application of integrated multi-omics technologies, including genomics, supported by AI-assisted facial and Gestalt recognition (**WP3**).
4) Install decision making support using multi-attribute analysis to optimize diagnostic and therapeutic choices enabling personalized optimal trajectories for RD-individuals, using a combined retrospective and prospective longitudinal study design (**WP4**).
5) Facilitate implementation of innovative and affordable RD-therapies via non-for-profit spin-offs aiming at in-house development by initiating a specialized platform involving relevant disciplines (ethics, legislation, entrepreneurship) (**WP5**).

Further fostering student engagement in research activities

In all aims, students from EMC-TUD-EUR will be extensively involved (**section-2c/d**). Bridging of student-participants of RD-IC and University programs will be optimized by research opportunities for students within the RD-IC (i.e., internships, papers, bachelor/master's theses), an annual open-research day (scientist-speed-dating sessions, (poster)presentations), introducing students to the multiple disciplines covered by RD-IC (**section-2d**), young investigator awards, and mentoring in research and grant writing.

Feasibility

Aims and deliverables of WPs are feasible regarding necessary **staff**, **budget, infrastructure**. Due to shared positions between disciplines (**Figure-4**), deliverables are attainable as tasks and are inter-changeable.

Plans to attracting external funding: **section-2c/d.**
Possible risks and mitigation strategies: **section-2c.**

3.c Work plan (1499 of 1998 words)

*Elaborate on the **long-term work plan**, including, if applicable, a specification of required facilities and a **data-and infrastructure** plan. Pay attention to overall coherence of the work plan.*

Figure-6 shows workplan and alignment with our goals.

In the WPs described, especially in WP2-5, we will focus on selected RD-groups with diverse phenotypes: hemophilia, sickle cell disease and Netherton syndrome (WP2: patient-journeys); neurodevelopmental delay, specific metabolic diseases and bone marrow failure syndromes/bleeding of unknown cause (WP3-4: timing, choice of diagnostic techniques difficult; WP4-5: therapeutic options lacking)[1]. Importantly, this will allow identifications of innovations in all WPs, which can be extrapolated to a spectrum of RDs, by WP1 (our communication hub), for > five years of the Flagship.

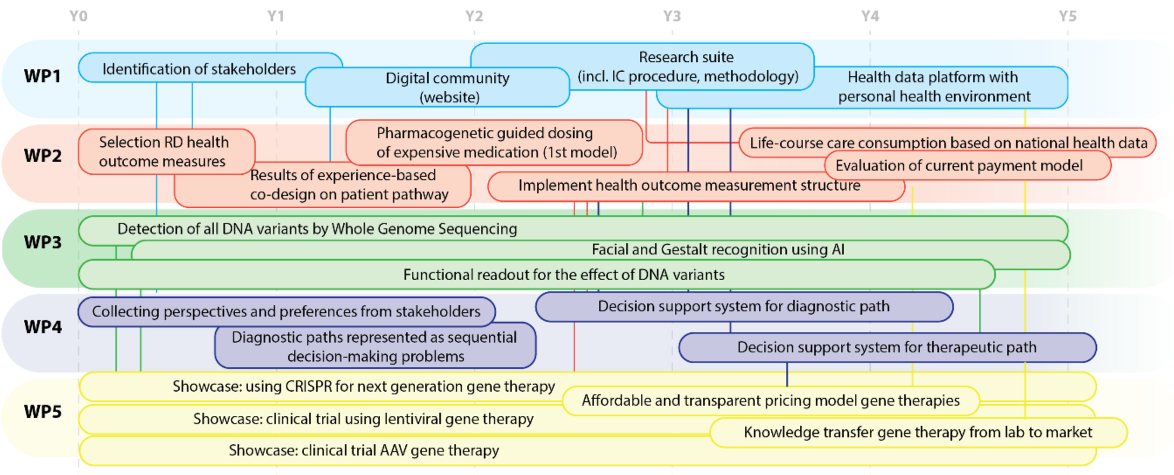

***Figure-6.** GANTT chart on timing of deliverables of the RD-IC Flagship program*

WP1: Rare disease community and data infrastructure
Leads/EMC: Pasmans, Schaaf/with Research suite EMC, Riedijk, Wijnen; EUR: Hulst, Uyl, TUD: Reinders
Aim: Improve ***communication and connections*** in the RD-community for all stakeholders to ensure findability and ***access to care/expertise and data***.

Objective/Unmet need		Tasks	Milestones	Deliverables
1	National RD-initiatives are insufficiently aligned; Lack of findable RD-expertise; stakeholders cannot find RD-individuals.	Identify and team-up with all (inter)national stakeholders, initiatives (M14-60)	-Connections/ partnerships with all national initiatives (M14) -Regular involvement at national RD-partner meetings (M24)	-Digital platform (M18) -Formation of RD-think tank
2	RD-individuals and RD-scientists cannot access health data (clinical/laboratory data, PROs/PREMS for healthcare, research)	Connect with (inter)national data-initiatives (HEALTH-RI, H2O)	-Collaboration-contracts (M12)	-RD-health data platform/Personal Health Record according to *MedMij* standards[2] (M60)
		Organize data-privacy issues, informed consent, access conditions/protection	-Informed consent at door of RD-IC	-RD-research suite according to FAIR principles (M42), allowing longitudinal follow-up and data-integration of RD-individuals with Generation R/ERGO population studies as example
		Construct RD-data-platform	-Pilot data-platform	

WP2: Improving RD care pathways; Innovating health care and empowering RD-individuals

Leads/EUR: Ahaus; EMC: Cnossen, Schappin; TUD: Albayrak

Aim: Improve ***RD-care pathways*** to become cost-effective and inclusive, with attention to health literacy and ethnic diversity of RD-individuals, by applying data-driven research including population based prognostic algorithms, experience-based co-design, integrating WP3/WP4/WP5 innovations.

	Objective/unmet need	Tasks	Milestones	Deliverables
1	RD care pathway is fragmented and incomplete.	Systematically collect health outcomes (biomarkers, clinical/patient-reported outcomes, patient-experience) efficiently	-Selection of RD health outcome measures (M12) -Identification of best ICT systems in electronic health records to actualize data-collection (M24)	-Digital system in which health outcomes are measured systematically (M48)
		Use RD-health data platform to collect and analyse data (WP1)	First analysis of health outcomes hemophilia, sickle cell disease (M50)	-2 manuscripts (M60)
		Apply experience-based co-design, patient/parent-journeys to improve care pathways, with attention to mental health/health literacy/ethnic diversity.	Patient-journeys for 4 specified-RDs in hemophilia, Netherton, undiagnosed neurodevelopmental delay (M24)	-2 RD-generic modules that facilitate organization of patient-centred disease-specific health journeys, based on best practices (M60)
		Shape care pathways towards developmental and life-course perspective, including transition to adult care	Patient-journeys for 4 specified-RDs with focus on transition (M36)	-Generic module, specifically for young adults in transition (M36)
		Social sciences study on implementation of	Concept report (M48)	-3 manuscripts (M60)

		innovations in care pathways.		
		Use population health data to create algorithms that identify RD and predict RD diagnoses based on care consumption	Identification of 4 RDs (M36)	-Prediction of 2 RDs (M60)
2	High treatment costs; payment models do not suffice.	Improve efficacy of treatment; decrease spillage by applying pharmacokinetic-guided dosing of expensive medication.	Data-collection complete hemophilia/von Willebrand cohort (n=50) (M30)	-2 population pharmokinetic-pharmocodynamic dosing models for medication in inherited bleeding disorders (M36)
		Investigate life-course care consumption, societal costs of specific RD-disease.	Concept report (M48)	-4 manuscripts (M60)
		Evaluate current payment models, build alternatives in collaboration with policy makers (Ministry, Healthcare Authority, Care Institute Netherlands) and insurance companies.	Concept report (M55)	-Report (M60)

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WP3: Integrated multi-omics technologies, supported by artificial intelligence assisted phenotype recognition to improve diagnostics (Figure-7)

Leads/EMC: van Ham, Barakat; TUD: van Gemert, Reinders; EUR: Uyl

Aim: Advance ***innovative RD-diagnostics*** for routine care, and employ AI-assisted image analysis to integrate outcomes with clinical features (WP4). We will increase RD-diagnostic yield from currently ~50% to >75% and reduce time to diagnosis (WP2), bringing more RD-individuals to the critical point for a fitting individualized care strategy (WP5).

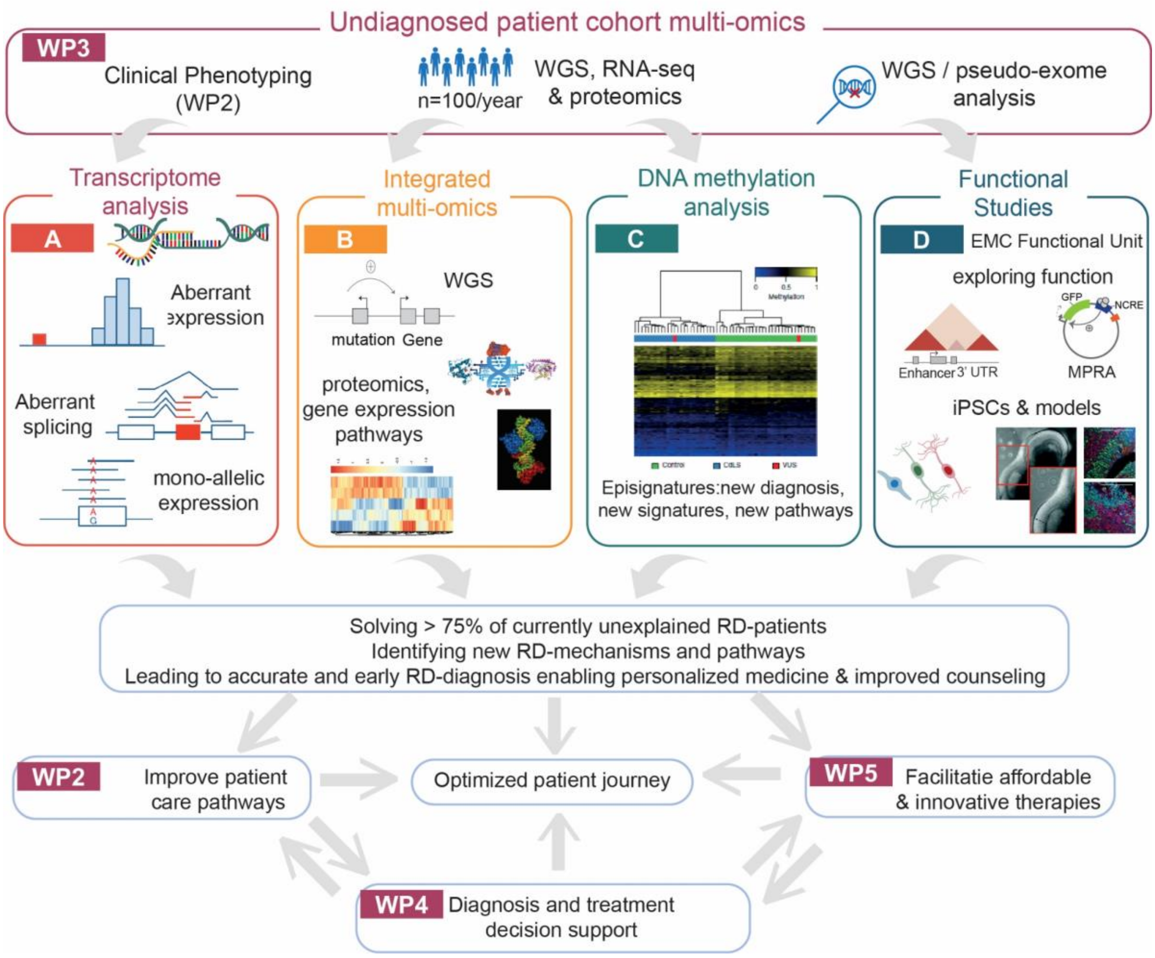

***Figure-7**: Multi-omics*

	Objective/unmet need	Tasks	Milestones	Deliverables
1	Increase RD-diagnostic yield as not all causative DNA-variants/causes are currently detected	Implement clinical WGS, RNA-seq, proteomics[3-10] (omics) as first-tier test in a prospective cohort (n=100 RD-individuals/year)	-Identify novel disease-causing variants and mechanisms in undiagnosed RD, including non-coding variants	-Diagnostic-omics pipelines cost-effectively implemented (with decision support (WP4) -Diagnostic yield to >75%: more diagnoses for RD-individuals (M55), facilitating personalized medicine (WP5) -2 manuscripts with findings (M60) -New disease mechanisms for future investigations
		Integrate analysis pipelines with artificial intelligence, prioritizing variants by synergy between WGS, RNA-seq, proteomics, fully exploiting multi-omics		
2	Reduce bias in Caucasian-dominated reference cohorts	Diversify reference cohorts by integrating genomics data from underrepresented ethnicities (UNITED consortium)	Increased assignment of DNA variants identifying false positive/negative cases	-Improved diagnostic yield in diverse Rotterdam population (M60)
3	Explore new RD-diagnostics technologies	Implement Med-Seq in RD-diagnostics, identifying methylation signatures in unexplained putative epigenetic disorders (Intellectual disability/growth-alterations, n=100/5years)	Identify chromatin disorder signatures	-Publications on RD-pathology -Implemented RD-diagnostics technology -2 manuscripts on methylation signatures (M48)
4	Improve automated facial/Gestalt recognition of syndromic disorders	Develop deep-neural network-based image	-Apply facial recognition in neurodevelopmen	-Systematic and low-cost screening tool

		recognition for genetic syndromes, using existing data (literature, EMC clinical genetics database). Pioneer computer vision-approaches based on few-shot learning to address non-balanced data (e.g., small-sized RD-cohorts)	tal disorders (pilot: chromatin-disorders).	to prioritize RD-individuals early that benefit from DNA-diagnostics, leading to earlier and improved diagnosis (M48) -Valorization of developed algorithms for routine care (M60)
5	Innovate technology in cost-effective manner	Perform health technology assessment/HTA to evaluate cost-efficient (WP2) and effective/smart use (WP4) of implemented technologies	Cost-effective use of -omics (when/which RD groups)	-Evidence-based precedent to implement omics-innovations in routine clinical RD-care -Expert-recommendation paper (M60)

WP4: Install diagnosis and treatment decision support using multi-attribute analysis.
Leads/TUD: Rezaei, Stankova, EUR: Irene Santi; EMC: Huidekoper, Veenma
Aim: Define ***alternative diagnostic paths considering perspectives of RD-individuals***, care providers and society to enable optimal trajectories for RD-individuals.

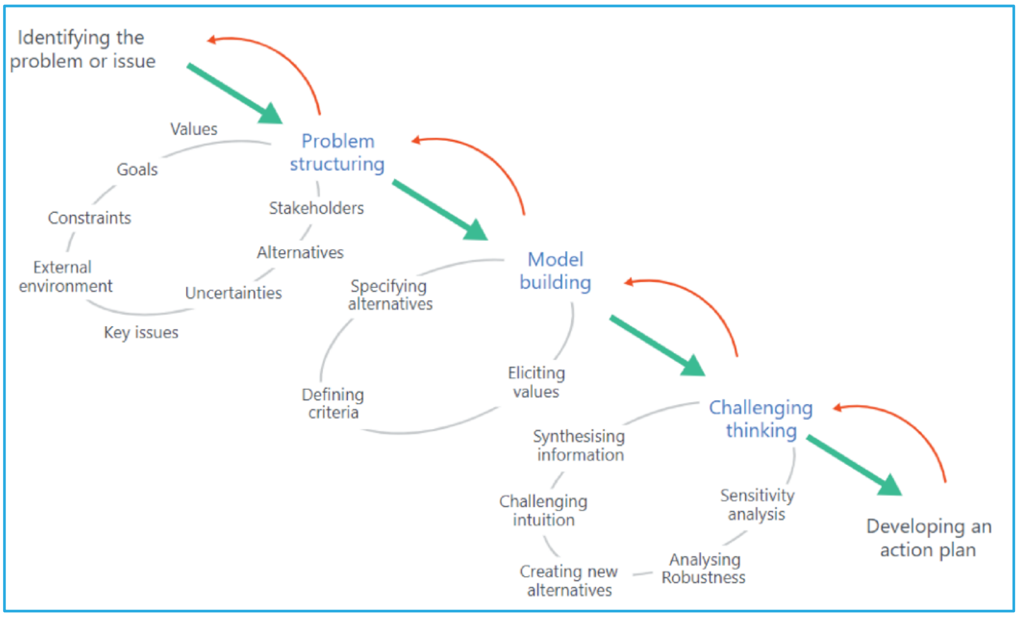

***Figure-8**: Sequential decision making.*

	Objective/unmet need	Tasks	Milestones	Deliverables
1	Diagnostic delay/low diagnostic yield are debit to low quality-of-life in RD-individuals.	Collect perspectives/preferences from stakeholders (WP1/2/3) (M<12)	-Conceptualize sequential decision-making problems, based on sequence of diagnostic events, maximizing probability of correct diagnosis in neurodevelopmental disorders (M24)	-Establishing/implementing algorithm detecting optimal diagnostic pathways, characterized by probability of diagnosis, total costs, total time, quality-of-life together with WP3 (M55) -2 manuscripts (M48)
2	Laborious, costly diagnostics.	Collect data from WP1 (RD-health data platform), WP2 (patient-journeys) and WP3 (diagnosis) (<M24)	-First analyses comparing costs of diagnostic pathway; establishment best practice for neurodevelopmental disorders (M50)	-2 manuscripts (M48)
3	Best diagnostics choice is complex, difficult for professionals.	Analyze data retrospective (via EMC clinical genetics) and prospective (WP3) RD-cohorts to develop decision support system using game theory (<M44)	-Test decision support system in different groups of stakeholders (M48)	-Decision support system using data form WP1/2/3 (M60)
4	Unclear which therapeutic approach should be prioritized according to outcomes, quality-of-life and costs.	Collect data from WP1 (RD-health data platform), WP2 (patient-journeys) and WP5 (therapies) (<M36)	-Concept report with guidelines for therapeutic choice in adults with hemophilia (M50)	-Algorithm to establish which therapeutic pathway is best concerning outcome(s), costs for hemophilia ∞WP5 (M60) -2 manuscripts (M60)

WP5: Facilitate implementation of innovative, affordable RD-therapies (gene, cellular, compound), fostering entrepreneurship **with integration of ethical considerations**
Leads/EMC: Pijnappel, Philipsen, Leebeek; TUD: van Grunsven; EUR: Jaspers
Aim: Establish an expert platform to build, retain, and provide expertise for users with four objectives. This will lead to a unique EMC Precision Medicine unit. Links: WP1/WP2 cost calculations for RD-therapies (EUR), Genomics in Society program (EMC/EUR).

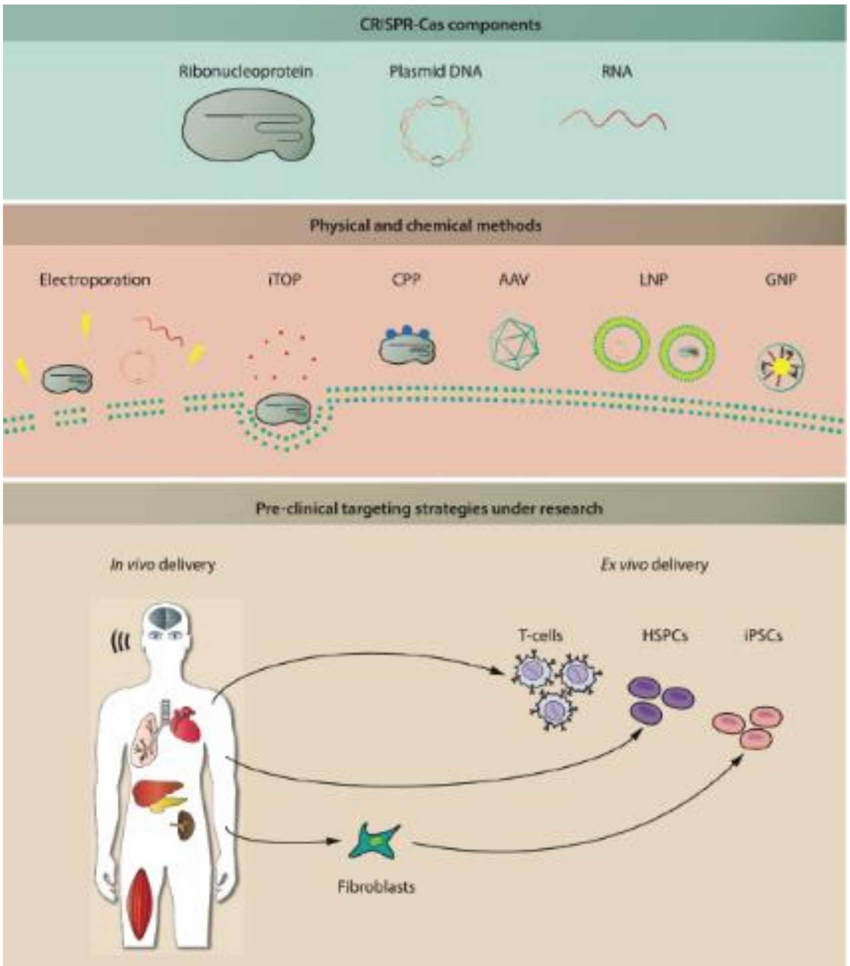

Figure-9: Overview of gene editing tools and strategies for clinical application[11-16]

	Objective/unmet need	Tasks	Milestones	Deliverables
1	Preclinical development of innovative therapies, with transition to clinical trials.	Preclinical development genetic therapies using CRISPR to correct sickle cell disease, hemophilia A/B (M60)	-Correction of sickle cell disease and hemophilia A/B achieved with >50% efficiency (M36, M60)	-Validated guide-RNAs for gene correction (M12, M36) -Delivery of RNPs to hematopoietic and endothelial cells with >50% targeted repair efficiency (M24, M48) -3 manuscripts (M60)
		Adeno-associated virus (AAV) episomal gene delivery for gene therapy of A/B and Pompe disease	-Phase 3 AAV-trial for hemophilia A/B completed (M60) -Phase 2 AAV-trial for Pompe disease completed (M48)	-Annual evaluation of AAV-trial for hemophilia A/B and Pompe (M12, 24, 36,48) -2 manuscripts M48, M60)
2	Design and performance of clinical trials		-Protocol for hemophilia B trial set up	Clinical trial set up for hemophilia (M36)
3	Entrepreneurship towards market authorization with the aim of in-house development via non-profit spin outs, taking ethical implications into account.	Education on business development of biologicals Qualitative research on ethical implications gene therapy in hemophilia	-First draft business plan/ gene therapy -Report ethics of gene therapy in hemophilia/ sickle cell disease	-Business plan (M36) -4 manuscripts
4	Establishment of a Erasmus MC Precision Medicine Unit	Acquire knowledge form required disciplines (a.o. legal, business, pharmacy) to develop such a unit	-Consent from board of directors for unit -Dedicated team and location in Erasmus MC	-Opening of precision medicine unit (M55)

Data Management Plan: described in Governance/**section-4d**.

3.d Alignment of the program with relevant national and international research agendas (69 of 1998 words)

Explain how the program aligns with or builds on national and international research agendas in its fields or, if it does not fit with these agendas, elaborate on what is the "high risk, high gain" objective.

This Flagship aligns RD-initiatives, including ERNs for RDs, the International RD-Research Consortium (IRDiRC), Solving the Unsolved Rare Diseases, and Care4Rare. The importance of improving care for RD-individuals has recently been emphasized by the Dutch parliament (motion 2021D18205). Furthermore, RD-IC aligns well within existing focus areas within EMC, such as the mainly pediatric RD-center and Child Brain Center. Potential agendas/calls for acquisition of external funding and funding strategies, see **sections-2c,d**.

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1. Consortium (25%) max 2000 words for 4.a to 4.d

- The Flagship program is driven and executed by a well-organized team of scientists from all three institutions and at least 4 departments or faculties in total. In exceptional cases, it is possible to form a Flagship program with only two leads from two institutions, but this should be clearly motivated (1)
- Members of the consortium have a track record of excellence in academic research, translation of scientific results into clinical application or societal impact (2)
- Members of the consortium have a significant track record in engaging and supervising BSc and MSc students in research projects (3)
- Members of the consortium have demonstrated to be able to attract external funding, thereby enhancing the programs sustainability after the 5 years of central funding (4)
- There is a demonstrable fit between the aims and objectives of the program and the expertise of the consortium members (5)
- End users (private and/or public sectors) are involved from an early start and team members are well-connected to external partners in the ecosystem (6)
- Governance should be clearly described and is organized in such a way that it facilitates collaboration and optimally makes use of the expertise of all participants (7)

4.a Track record of the consortium members (485 of 2000 words)

Elaborate on the ***consortium members' track record in academic research, translation of scientific results into clinical application or societal impact, BSc and MSc supervision and engagement*** *in research projects, ability to* ***attract external funding*** *after the 5-year central funding period.*

Consortium Scientific track record

This Flagship encompasses 23 core scientific members, 21 co-applicants, representing a diverse (inter)national community (21 female, 24 males) of both experienced and early career scientists from EMC-TUD-EUR, of which the majority has previously trained abroad. Together, the members have authored > 2500 peer-reviewed publications in fields relevant to this proposal, and have a track record of acquiring >€60 million external funding (*also see matching*), including personal fellowships (EMBO, Marie Skłodowska-Curie fellowships, Veni, Vidi, Vici, ERC, EUR and EMC Fellowships, with many early-career consortium researchers still eligible for future calls), and other competitive funding programs, including amongst others Horizon 2020, ZonMw/NWO TOP, NWO-NWA, NWO-Gravity, ZonMw-PSIDER. Societal impact is exemplified by the DNA dialogue funded by the Dutch Ministry of Health, and KNAW Funds “Science Communication: Appreciated!” Recent awards for consortium members include the “Zeldzame Engel award” (Pasmans, recognition RD-engagement) and an KNAW Early Career Award (Barakat, for genetic RD-research). Together, this shows the academic excellence and experience of the partners involved and supports the capacity to attract additional funding ensuring the Flagship sustainability.

Track record of translation of scientific results in clinical application and societal impact

RD-IC members have a longstanding track record of translating scientific findings to the clinical, have written more than 20 national evidence-based guidelines for RD-individuals[10] (Cnossen, de Graaff, Escher, Leebeek, Veenma, Dinkelman, Pasmans, Huidekoper, Galjaard, van Dooren) all 10 health professionals are advisors to various patient-organizations, the government and are (or were) board members of health professional associations, >8 use patient-reported outcomes in outpatient care, >2 have founded foundations to raise money for RD-individuals (Cnossen, Pasmans), and have been principal investigator >15 clinical drug trials. All are leader of their NFU RD-center and Theme Sophia focus area. Specialists in RD-diagnostics and genetics have identified >20 new genetic RD-syndromes[1-9] (leading to improved counseling, prevention, therapy, informed reproductive choices) and have developed and implemented >3 new laboratory technologies to optimize diagnosis[11-13]. Fundamental scientists are leading in their field and are on the bridge of developing >6 genetic therapies and >3 novel biochemical compounds to modify disease outcome.

Track record in engaging and supervising BSc and MSc students in research projects

RD-IC members have a longstanding experience in supervising >600 students at various levels (>18 members are BKO/UTQ certified), and are currently supervising jointly >70 BSc/ MSc-students, >100 PhD-students, >20 clinical residents/fellows and >50 postdoctoral researchers in their research groups. Amongst the Flagship members are faculty members of the MSc programs Medisign and Deep learning and Computer Vision MSc courses at TUD, MSc Healthcare Management and Health economics at EUR, and the MSc Molecular Medicine and creators (Riedijk, Adams, van Ham, Barakat) of the MSc Genomics in Society Masters at EMC. The latter MSc program (starting Sept. 2022) is specifically tailored to educate future policy makers and stakeholders that will deal with dialogues on genomics in society, and are therefore perfectly suited to align with the research initiatives within RD-CI.

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4.b Expertise of the consortium members in relation to the aims and objectives of the program (361 of 2000 words)

*Demonstrate the **fit between the aims and objectives of the program and the expertise of the consortium members**. Also describe the complementarity of the members.*

This Flagship aligns with the aims and objectives of the program when looking at the expertise and complementarity of all the consortium members together, including external partners (**section-1a**).

Theme	Disciplines	Expertise	Members
Improving health journeys; focus on research/ education	*RD-individuals and families*	Patient and family experiences, advocacy/ organizations	External partners: VSOP, Patiëntenfederatie,
	Medicine: medical RD-specialists e.g. hematology, metabolic diseases, neurodevelopmental disorders, neurology, clinical genetics, urology, IBD, pediatric surgery, dermatology, clinical pharmacology, radiology.	RD-care and research; life-course medicine (pediatric & adult perspective); genetic counseling; prenatal/ implementation techniques; sexuality; inherited anatomical disorders; ERN leader-ties to Europe; therapeutic drug monitoring	Cnossen[14-18], Huidekoper[19-23], Veenma[24-27], , de Graaff, Mancini[1-5], van Dooren[28-30], Escher[31-35], Dinkelman-Smit, Galjaard[36,37], Wijnen, Pasmans[38-40], Koch, Adams[41-44]. External partners: van der Ploeg[45-46] (heads RD-centers EMC), Mathijssen (lead ERNCranio)
	Social sciences (incl. paramedical RD-care)	Mental health care, psychosocial interventions, patient-journey, experience-based co-design, value-based healthcare, patient-reported outcomes, health care management, implementation science, communication	Ahaus[47-51], van de Bovenkamp[52-57], Riedijk[58-60], Schappin[61-64]
	Industrial design	Human-centred design, experience-based co-design, health journey, communication, outreach	Albayrak[65-67], Souhoka

	Ethics	Ethics education, philosophy of technology, biomedical ethics	Grunsven[68] <u>External partners:</u> Bredenoord
	*Epidemiology and data science**	Decision making, game theory; decision making in small RD samples; methodology and statistics; data storage; mathematics; pharmacometrics	Adams[41-44], Ahaus[49-51,69,70], Lingsma, Rezaei[71-74], Santi[75-79], Schappin, Stankova[80-84] Research suite/contacts: Pasmans, Schaaf, Koch/ Cnossen. <u>External partners:</u> PHARMO
	*Health economics**	Health Technology Assessment	Ahaus, Uyl[85-86], Klein, Santi, Versteegh[87-91], Jaspers,
	Teaching/ education	Innovative interdisciplinary teaching modules	Riedijk, Adams, Souhoka
	*Health law**	Legal/ privacy issues	Hulst[92-93], Buijsen, den Exter
	*Societal and research partners**	Public and private investors; Governmental organizations (ZiN, ZonMW)	<u>External partners:</u> Van Meeteren, others approached
Technology-supported transitions in health(care)	*RD-genetics and diagnostics*	Genomics/ multi-omics, Med-Seq, proteomics, ICT/ computer sciences, artificial intelligence/ deep learning; image recognition	Barakat[6,7,10-12], van Ham[6,9,94-96], Hoefsloot[97-98], Ruijter[99-103], Gribnau[13,104,105], Demmers[106-108], van Gemert[109-110], Reinders[111-114], Mulugeta[12,115-116]
	RD-therapies	Molecular genetics; molecular biology; cell biology; gene editing; biochemistry (compounds); entrepreneurship/ business; ethics of	Leebeek, Philipsen[117-121], Pijnappel[122-126], Schaaf, Jaspers, van Grunsven, Roeser

		technology; public opinion	
Human Centered Technology and AI for Health	*Data-science*; see also health journey	Artificial intelligence	Reinders[111-114], van Gemert[109-110], Mulugeta[12,115-116]
Fundamentals of health and disease	*Molecular genetics and molecular techniques;* see technology-		

Figure-10. *Kick-off meeting Rare Disease Innovation Center, Boijmans Depot, Friday February 25th 2022*

4.c Involvement of partners (160 of 2000 words)

Indicate which partners are involved, how and in what stage they are involved, as well as how well team members are connected to the external partners in the ecosystem.

Our external partners in **Table below** demonstrate the connections and complementarity within our proposal (**section-4b**)

Name of partner	Anticipated Input	WP involvement	Connection with team members
VSOP	RD-individual voice, alignment with other national initiatives	1/2	EMC (a.o. Cnossen, Pasmans)
Patientenfederatie Nederland*	RD-individual voice, alignment with other national initiatives	1/2	EMC (a.o. Cnossen, Pasmans)
RD-centers/ERNs, later EURORDIS (patient-society)	Networking, knowledge and data-sharing,	1-5	EMC (a.o. Wijnen, Mathijssen, van der Ploeg)
TKI-Health Holland	Public/ private partnerships and funding	1-5	All RD-professionals
PHARMO Institute	Sharing knowledge and public health data	1/2/4	EMC (Schappin, Pasmans)
Patient-organizations, amongst others Stichting12q	RD-individual and family voice, increase public awareness, funding	1-5	All RD-professionals

Professional/scientific associations	Networking, knowledge/data-sharing	1-5	All RD-professionals and scientists
ZiN* Zorginstituut Nederland	Shared interest in cost-efficiency issues, reimbursement/expensive medication	1/2	EMC (Cnossen, Pasmans)
ZonMW* ZonMw	Research agenda for RD	1-5	EMC (Pasmans)
Pharos* PHAROS EXPERTISECENTRUM GEZONDHEIDSVERSCHILLEN	Health literacy expert	1-5	EMC (Cnossen, Schappin, Pasmans)

**approached and are enthusiastic, but cannot commit (yet) due to managerial issues*

4.d Governance, organization and collaboration strategy of the consortium (994 words)

Please describe how the consortium is ***organized to facilitate collaboration and make optimal use of the expertise of all participants.***

Governance
The governance of the flagship is organized at three management levels (**Figure-11**):

Level 1: Action
The execution of each WP is managed by **WP representatives from each of the three universities** to accelerate interdisciplinary synergy (=**WP leads; named in section-3c**).

Level 2: Management
As this Flagship is a merge between two former flagship pre-proposals, it is jointly led by three leads with different backgrounds from the EMC/TUD, together with six co-leads from all three universities and per WP a lead from EMC-TUD-EUR, if not already a lead or co-lead. This scientific core team acts as the **Executive board** (Leads: Cnossen, Albayrak, Barakat; Co-leads: van Ham, Schappin, Ahaus, van Gemert, Rezaei, Uyl; Applicants not already lead or co-lead: WP1: Hulst, Wijnen, Pasmans, Schaaf, Riedijk; WP3: Reinders; WP4: Huidekoper, Santi, Veenma, Stankova; WP5: Pijnappel, Leebeek, Philipsen, Grunsven, Jaspers; **Figure-12**). It is supported by a program coordinator and communication officer for organizational activities. The Executive board monitors scientific progress and addresses issues. It consists of a diverse group of female and male multidisciplinary scientists, with relevant expertise, both experienced researchers as well as

early career scientists, reflecting our ambition to establish a future proof sustainable research environment driving the convergence. Program coordinator and leads are responsible for day-to-day management including monitoring of milestones and deliverables, knowledge transfer and utilization, contractual, legal, financial, and administrative affairs, and organization of meetings.

Level 3: Decision and advice
The Supervisory board is the ultimate decision-making entity and addresses pressing issues that affect the Flagship. The **Supervisory board** consists of a representative from each institute (EMC: Cnossen/Barakat; TUD: Albayrak; EUR: Ahaus) and is authorized to make binding decisions on behalf of his/her party.

The **Advisory board** gives expert advice to the Flagship on its strategy and consists of advisors with special expertise, representatives of important stakeholders/public and private partners, most importantly patient-organizations (Patiëntenkoepel zeldzame en genetische aandoeningen/VSOP; Patiëntenfederatie Nederland), and international experts (**section-4c**, **section-1a**). The NFU RD-centers in the EMC/ERNs are represented by van der Ploeg and Mathijssen. A **Patient panel** will be installed to give critical feedback on interventions and choices. According to national guidelines budget is allocated, to compensate for time and involvement.

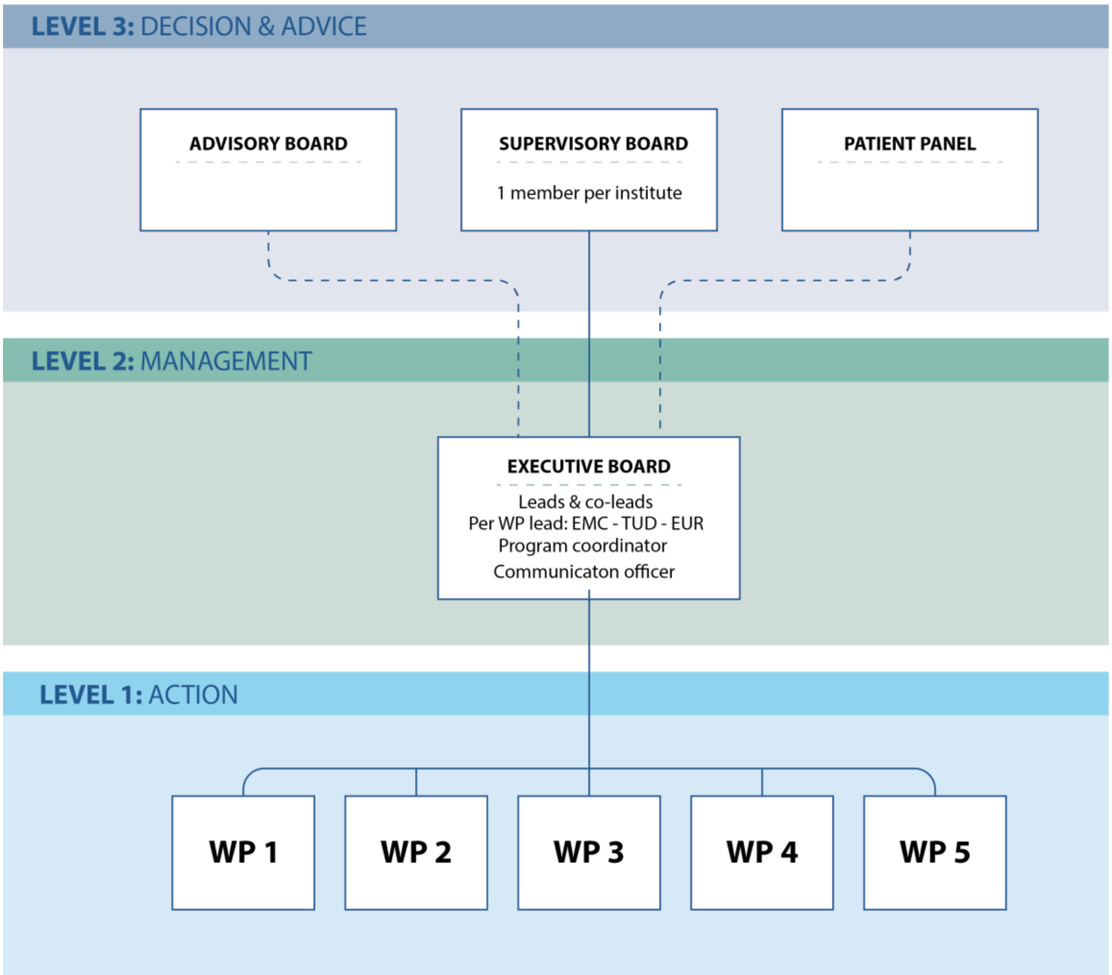

Figure-11. *Governance of Rare Disease Innovation Center initiative*

Figure-12. *Executive board of the* ***Rare Disease Innovation Center initiative***

Organization

The Executive board will have teleconference meetings every month. WPs themselves will meet at least monthly. RD-research (hybrid) meetings for professionals and scientists from all institutes in the Flagship will take place every month. An annual two-day/later 3-day conference (**section-2d**) will be organized focusing both on ongoing results as well as new initiatives and funding opportunities. The EMC NFU-RD-centers will meet every four months to share novel developments. An overview of planned meetings is provided in **Table below.**

Body		**Meetings #/year**
Supervisory board		1
Executive board		12 (=monthly)
Program coordinator, Communication coordinator and leads		50 (=weekly)
RD-IC research meetings*		12
Annual two-day conference*#		1
Advisory board		≥1
Patient panel		≥1
NFU-RD-centers/ERN		3

* All researchers from three institutes invited as well as interested BSc/BA and MSc/MA-students. #A small fee will be asked for organizational costs, but free of charge for students.

Communication strategy: Communication will be supported by a coordinator.

Data management
Flagship members will have knowledge of and comply to institutional and (inter)national research codes (Institutional Research Codes, European Code of Conduct for Research Integrity). Data will be handled practicing the H2020 FAIR principles and aligned to recommendations of the Health-RI initiative. All results are documented in notebooks and digital data are stored in a long-term archive on password-protected network servers, supported by the Research Suite, with regular back-up. In accordance with the Netherlands Code of Conduct for Scientific Practice, raw data is stored for 10 years. Primary patient materials will be securely stored at centralized (e.g., Erasmus MC Biobank, Vriesveem) and approved departmental storage sites, ensuring protected access.

All patient related work has been or will be approved by the EMC IRB and will comply with all relevant legislation (e.g., Dutch Personal Data Protection Act (WBP), Dutch Medical Research involving Human Subjects Act (WMO), European regulations including the GDPR (EC 2016/679)), in line with the declaration of Helsinki. Detailed procedures for data handling, protection and pseudo-anonymization will be described in each IRB protocol. All patients/legal guardians will give written informed consent. To ensure compliance with regulations, we will be supported by Data Management Offices of the institutions, local data managers and PaNaMa, the research project management system that supports registration and administrative processes of human-related research at Erasmus MC.

To make the output of this Flagship **f**indable, **a**ccessible, **i**nteroperable and **r**eusable (FAIR) for all relevant stakeholders, ensuring long-term sustainable, we will follow the following strategies:

- Appropriate European infrastructure (endorsed by ELIXIR) will be used to ensure patient data confidentiality and only fully anonymized datasets will be deposited in open access data repositories. Sequence data will be submitted to the European Genome-Phenome Archive (EGA), which provides managed access to sensitive data for approved researchers. Selected processed results will be made available via ArrayExpress/Gene Expression Omnibus (e.g., data which no longer contains primary DNA sequences and is fully anonymized: gene expression data).
- Containerized versions of data processing pipelines developed will be submitted to biocontainers.org to enable effective reuse and advanced analysis code will be shared via GitHub.
- Scientific results will be submitted to international high-impact, peer-reviewed journals, under open-access to enable broad visibility. We will make manuscripts available as pre-prints on BioRxiv, to ensure fast dissemination.
- Our findings will be presented at various international scientific meetings, (e.g., Annual meetings of European and American Society of Human Genetics), to increase awareness in the RD-community.

For third party access to primary data prior to publication, specific procedures will be developed in the Research Collaboration Agreement, and access will be subject to material transfer agreements (MTAs) and data transfer agreements (DTAs) that will be developed together with the Technology Transfer Offices of our institutes. Data ownership and accessibility will always be discussed with patient panel to ensure the RD-individual perspective.

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Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

Annex I - Budget table (including the external funding table)

Budget formulier: Convergentie Flagship Call

Flagship:	Rare Disease Innovation Center (RD-IC) initiative: Converging stakeholders and sciences to propel the health journey	
Flagship acronym:	Rare Disease Innovation Center (RD-IC)	
Naam Penvoerder:	Dr. M.H. Cnossen	
Startdatum en looptijd van het project:	Startdatum 01-09-2022	Looptijd: 5 jaar
Totaal project budget:	€	4.610.379,60

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1. **Loonkosten:**

Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten	2023		2024		2025		2026		2027		Totaal		Totaal per partner
								in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																				
EUR / TU Delft	PostDoc1	0,80	2 Years		PostDoc University	€ 5.222	€ 150.400	€ 75.200,00								€ 75.200,00		€ 150.400		€ 150.400,00
Erasmus MC	Medical PhD student1	1,00	4 Years		Medical PhD student	€ 4.000	€ 288.000	€ 72.000,00		€ 72.000		€ 72.000		€ 72.000				€ 288.000		€ 288.000,00
Erasmus MC	PostDoc2	0,80	2 Years		Schaal 10	€ 4.802	€ 138.298	€ 69.148,80		€ 69.149								€ 138.298		€ 138.297,60
TU Delft	PhD student2	1,00	3 Years		PhD student University	€ 3.189	€ 172.213					€ 57.404		€ 57.404		€ 57.404,25		€ 172.213		€ 172.212,75
Erasmus MC	PhD student3	1,00	3 Years		OIO	€ 3.333	€ 180.000					€ 60.000		€ 60.000		€ 60.000,00		€ 180.000		€ 180.000,00
EUR	PhD student1	1,00	4 Years		PhD student University	€ 3.189	€ 229.617			€ 57.404		€ 57.404		€ 57.404		€ 57.404,25		€ 229.617		€ 229.617,00
TU Delft	PhD student4	1,00	3 Years		PhD student University	€ 3.189	€ 172.213	€ 57.404,25		€ 57.404		€ 57.404						€ 172.213		€ 172.212,75
EUR	PhD student5	1,00	4 Years		PhD student University	€ 3.189	€ 229.617			€ 57.404		€ 57.404		€ 57.404		€ 57.404,25		€ 229.617		€ 229.617,00
Erasmus MC	Program manager	0,40	5 Years		Schaal 10	€ 4.802	€ 172.872	€ 34.574,40		€ 34.574		€ 34.574		€ 34.574		€ 34.574,40		€ 172.872		€ 172.872,00
TU Delft	Program secretary	0,40	2 Years		Secretary University	€ 3.422	€ 49.277	€ 24.638,40		€ 24.638								€ 49.277		€ 49.276,80
Erasmus MC	PhD student	1,00	2 Years		OIO	€ 3.333	€ 120.000	€ 60.000,00		€ 60.000								€ 120.000		€ 120.000,00
Erasmus MC	PostDoc	0,30	2 Years		Schaal 10	€ 4.802	€ 51.862	€ 25.930,80		€ 25.931								€ 51.862		€ 51.861,60
Erasmus MC	Technician	0,50	2 Years		Technician	€ 3.333	€ 60.000	€ 30.000,00		€ 30.000								€ 60.000		€ 60.000,00
Erasmus MC	PhD student	0,50	2 Years		OIO	€ 3.333	€ 60.000	€ 30.000,00		€ 30.000								€ 60.000		€ 60.000,00
Erasmus MC	PhD student	0,50	2 Years		OIO	€ 3.333	€ 60.000	€ 30.000,00		€ 30.000								€ 60.000		€ 60.000,00
Erasmus MC	PhD student	1,00	2 Years		OIO	€ 3.333	€ 120.000	€ 60.000,00		€ 60.000								€ 120.000		€ 120.000,00
Erasmus MC	PhD student	1,00	2,33 Years		OIO	€ 3.333	€ 140.000	€ 60.000,00		€ 60.000		€ 20.000						€ 140.000		€ 140.000,00
Erasmus MC	PhD student	0,50	2 Years		OIO	€ 3.333	€ 60.000	€ 30.000,00		€ 30.000								€ 60.000		€ 60.000,00
Erasmus MC	PostDoc	0,17	2 Years		Schaal 10	€ 4.802	€ 28.932	€ 14.466,00		€ 14.466								€ 28.932		€ 28.932,00
TU Delft	PostDoc	0,40	1 Year		PostDoc University	€ 5.222	€ 37.880	€ 37.880,00										€ 37.880		€ 37.880,00
TU Delft	PhD student	0,49	2 Years		PhD student University	€ 3.189	€ 55.834	€ 27.917,00		€ 27.917,00								€ 55.834		€ 55.834,00
Erasmus MC	PhD student	0,50	4 Years		OIO	€ 3.333	€ 120.000	€ 30.000,00		€ 30.000,00		€ 30.000,00		€ 30.000,00				€ 120.000		€ 120.000,00
Erasmus MC	PostDoc	1,00	2 Years		Schaal 10	€ 4.802	€ 172.872	€ 86.436,00		€ 86.436								€ 172.872		€ 172.872,00
Erasmus MC	PhD student	1,00	2 Years		OIO	€ 3.333	€ 120.000	€ 60.000,00		€ 60.000								€ 120.000		€ 120.000,00
Erasmus MC	PhD student	1,00	2 Years		OIO	€ 3.333	€ 120.000	€ 60.000,00		€ 60.000								€ 120.000		€ 120.000,00
Unknown	PostDocs Roadmap funding possibilities	1,00	3 Years		PostDoc University	€ 5.222	€ 282.000					€ 94.000		€ 94.000		€ 94.000,00		€ 282.000		€ 282.000,00
Unknown	PhD students Roadmap funding possibilities	4,00	3 Years		OIO	€ 3.333	€ 720.000					€ 240.000		€ 240.000		€ 240.000,00		€ 720.000		€ 720.000,00
Unknown	Medical PhD students	1,00	3 Years		Medical PhD student	€ 4.000	€ 216.000					€ 72.000		€ 72.000		€ 72.000,00		€ 216.000		€ 216.000,00
						Totaal	€ 4.327.886												Totaal loonkosten	€ 4.327.886

2. **Kosten van materialen en hulpmiddelen:**

Partner die de kosten maakt	Materiaal / hulpmiddel		Kosten	2023		2024		2025		2026		2027		Totaal		Totaal per partner
				in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																
Erasmus MC	Material costs		€ 20.000	€ 7.500,00		€ 3.000				€ 6.500		€ 3.000,00		€ 20.000		€ 20.000,00
EUR	Material costs		€ 20.000	€ 7.500,00		€ 3.000				€ 6.500		€ 3.000,00		€ 20.000		€ 20.000,00
Erasmus MC	Material costs		€ 37.161	€ 17.267,08		€ 4.298				€ 13.798		€ 1.797,88		€ 37.161		€ 37.160,70
TU Delft	Material costs		€ 20.000	€ 7.500,00		€ 3.000				€ 6.500		€ 3.000,00		€ 20.000		€ 20.000,00
Erasmus MC	Material costs		€ 37.161	€ 17.267,08		€ 4.298				€ 13.798		€ 1.797,88		€ 37.161		€ 37.160,70
Erasmus MC	Patient involvement		€ 21.809	€ 5.000,00		€ 5.000		€ 3.808,60		€ 5.000		€ 3.000,00		€ 21.809		€ 21.808,60
Erasmus MC	Valorisation strategy		€ 25.000									€ 25.000,00		€ 25.000		€ 25.000,00
Erasmus MC	Data storage		€ 5.000									€ 5.000,00		€ 5.000		€ 5.000,00
Erasmus MC	Consumables		€ 55.000	€ 13.750,00		€ 13.750,00		€ 13.750,00		€ 13.750,00				€ 55.000		€ 55.000,00
Erasmus MC	Consumables		€ 10.000	€ 5.000,00		€ 5.000								€ 10.000		€ 10.000,00
																€ 0,00
		Totaal	€ 251.130												Totaal kosten materialen en hulpmiddelen	€ 251.130

3. **Kosten van gebruik van machines en apparatuur**

Partner die de kosten maakt	Machine / apparatuur			Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	2023		2024		2025		2026		2027		Totaal		Totaal per partner
							in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
																			€ 0,00
																			€ 0,00
																			€ 0,00
																			€ 0,00
					Totaal	€ 0												Totaal kosten machines en apparatuur	€ 0

4. **Aan derden verschuldigde kosten**

Partner die de kosten maakt	Naam derde, omschrijving kosten		Kosten	2023		2024		2025		2026		2027		Totaal		Totaal per partner
				in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																
																€ 0,00
																€ 0,00
																€ 0,00
																€ 0,00
		Totaal	€ 0												Totaal kosten derden	€ 0

5. **Publicatie, reis- en verblijfkosten**

Partner die de kosten maakt	Naam derde, omschrijving kosten		Kosten	2023		2024		2025		2026		2027		Totaal		Totaal per partner
				in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																
Erasmus MC	Meeting costs		€ 16.364	€ 5.000,00		€ 4.830				€ 4.117		€ 2.417,10		€ 16.364		€ 16.364,10
Erasmus MC	Publication costs		€ 15.000							€ 5.000		€ 10.000,00		€ 15.000		€ 15.000,00
																€ 0,00
																€ 0,00
		Totaal	€ 31.364												Totaal kosten derden	€ 31.364

Totale kosten	Kosten	Totaal per partner
Totale kosten	€ 4.610.380	€ 4.610.380

FINANCIERING - Bijdragen per partner per jaar

in Euro	2023 in cash	2023 in kind	2024 in cash	2024 in kind	2025 in cash	2025 in kind	2026 in cash	2026 in kind	2027 in cash	2027 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties													
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
Totale bijdrage onderzoeksorganisaties	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
Ondernemingen													
Roche Diagnostics Nederland	€ 60.000		€ 60.000		€ 20.000						€ 140.000	€ 0	€ 140.000
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
Totale bijdrage ondernemingen	€ 60.000	€ 0	€ 60.000	€ 0	€ 20.000	€ 0	€ 0	€ 0	€ 0	€ 0	€ 140.000	€ 0	€ 140.000
Overige (private) partners													
Stichting12q Research Fellowship	€ 65.000		€ 65.000								€ 130.000	€ 0	€ 130.000
CURE Epilepsy Award / EpilepsieNL Grant	€ 55.931		€ 55.931								€ 111.862	€ 0	€ 111.862
NWO/ENW KLEIN2	€ 65.797		€ 27.917								€ 93.714	€ 0	€ 93.714
SYMPHONY NWO-NWA	€ 30.000		€ 30.000								€ 60.000	€ 0	€ 60.000
SYMPHONY WP8	€ 30.000		€ 30.000								€ 60.000	€ 0	€ 60.000
SYMPHONY WP6	€ 60.000		€ 60.000								€ 120.000	€ 0	€ 120.000
Netherton	€ 44.466		€ 44.466		€ 0		€ 0		€ 0		€ 88.932	€ 0	€ 88.932
Tracer ZonMw	€ 43.750		€ 43.750		€ 43.750		€ 43.750				€ 175.000	€ 0	€ 175.000
Prinses Beatrix Spierfonds Pijnappel	€ 206.436		€ 206.436		€ 0		€ 0		€ 0		€ 412.872	€ 0	€ 412.872
Roadmap funding possibilities					€ 406.000		€ 406.000		€ 406.000		€ 1.218.000	€ 0	€ 1.218.000
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
Totale overige bijdrage	€ 601.380	€ 0	€ 563.500	€ 0	€ 449.750	€ 0	€ 449.750	€ 0	€ 406.000	€ 0	€ 2.470.380	€ 0	€ 2.470.380
Flagship													
Flagship	€ 400.000		€ 400.000		€ 400.000		€ 400.000		€ 400.000		€ 2.000.000	€ 0	€ 2.000.000
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
Totaal subsidies	€ 400.000		€ 400.000	€ 0	€ 400.000	€ 0	€ 400.000	€ 0	€ 400.000	€ 0	€ 2.000.000	€ 0	€ 2.000.000
	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			Totaal incl. Flagship
Totaal budget per jaar	€ 1.061.380	€ 0	€ 1.023.500	€ 0	€ 869.750	€ 0	€ 849.750	€ 0	€ 806.000	€ 0			€ 4.610.380
													Funding gap
													-
Totaal budget	€ 1.061.380		€ 1.023.500		€ 869.750		€ 849.750		€ 806.000		€ 4.610.380		

Annex II - Letters of intent and letters of commitment

Erasmus MC
University Medical Center Rotterdam

Pediatrics
Prof. E.H.H.M. Rings, MD, PhD
Prof. A.M.C. van Rossum, MD, PhD

Prof.dr. S. Sleijfer
voorzitter Raad van Bestuur Erasmus MC

Internal postal address **Sp 2471**
Date March 11, 2022

Postal address
P.O. box 2060
3000 CB Rotterdam, NL

Street address
Wytemaweg 80
3015 CN , Rotterdam, NL

Car Park
Westzeedijk 361
3015 AA Rotterdam

LETTER OF INTENT *for the* ***Rare Disease Innovation Center Flagship PROJECT***

Dear Professor S. Sleijfer,

I, **Prof. Edmond H.H.M. Rings**, in my capacity of head of the department of Pediatrics at Erasmus University Medical Center, Erasmus MC Sophia Children's Hospital hereby confirm that *Erasmus University Medical Center* Rotterdam intends to contribute to the flagship *the Rare Disease Innovation Center*, as applied for by the main applicant, **Dr. Marjon H. Cnossen, pediatric hematologist at Erasmus MC Sophia Children's Hospital.**

- Erasmus University Medical Center intends to contribute **€ 140.000,00** in cash towards the project costs in accordance with the budget in the project proposal and budget from.

Leading to a total contribution of € **140.000,00** which will function as co-financing for the first two years of the flagship.

Both contributions are from a project that aligns with the flagship aims. This project was jointly initiated by Erasmus MC, Erasmus University Rotterdam and TU Delft and is called *Partitura.* It aims to construct the patient journey for young children (0-8 years of age) with

Erasmus MC
Universitair Medisch Centrum Rotterdam

an inherited rare bleeding disorder.

Yours sincerely,

Prof. Dr. Edmond H.H.M. Rings
Head of Department of Pediatrics
Erasmus MC Sophia Children's Hospital

Date: March 8, 2022

Other involved project leaders of *Partitura,* agree with this in cash and in kind contribution.

Prof. Dr. Kees Ahaus
Erasmus School of Health Policy & Management Health Services Management & Organisation (HSMO)/ Erasmus University Rotterdam
Date: March 11, 2022

Dr. Armagan Albayrak
Assistant professor Applied Ergonomics & Design, Technology University of Delft Technology
Date:

Dr. Marjon H. Cnossen
Associate professor Pediatric Hematology
Erasmus MC Sophia Children's Hospital
Date: March 11, 2022

Prof. Dr. Hester Lingsma
Head Medical Decision making
Erasmus MC
Date: March 11, 2022

Erasmus MC
Universitair Medisch Centrum Rotterdam

Prof. dr. E.H.H.M. Rings
Dr. Molewaterplein 40
3015 GD Rotterdam

March 11th, 2022

LETTER OF INTENT
for the
Rare Disease Innovation Center

Dear Prof. dr. Rings,

I, Prof. dr. Tamar Nijsten, in my capacity of head of the Department of Dermatology at Erasmus MC hereby confirm that Erasmus MC intends to contribute to the Flagship Rare Disease Innovation Center, as applied for by the main applicant, Marjon Cnossen, associate professor Pediatric Hematology at Erasmus MC.

Erasmus MC intends to provide an in-kind contribution of 0.5 fte PhD and 0.1 fte postdoc on Netherton Syndrome, from 01-09-2022 to 31-08-2024, representing a monetary value of € 88,932.00 and further detailed in the project proposal and budget form.

Yours sincerely,

namens/on behalf of

Name: Prof. dr. T.E.C. Nijsten

Position: Head of Department of Dermatology

Date: 11-03-2022

Erasmus MC
University Medical Center Rotterdam

Sophia Kinderziekenhuis
afdeling Kindergeneeskunde
Prof.dr. E.H.H.M. Rings/ Prof.dr. A.M.C. van Rossum

Prof. dr. S. Sleijfer
Voorzitter en decaan Erasmus MC

Internal postal address: SP 2471
Date: March 11th, 2022

LETTER OF INTENT** for the **Rare Disease Innovation Center Flagship PROJECT

Dear Professor S. Sleijfer,

I, **Prof. Dr. Edmond Rings**, in my capacity of head of the department of Pediatrics at Erasmus University Medical Center, Erasmus MC Sophia Children's Hospital hereby confirm that *Erasmus University Medical Center* Rotterdam intends to contribute to the flagship the Rare Disease Innovation Center, as applied for by the main applicant, **Dr. Marjon H. Cnossen, pediatric hematologist at Erasmus MC Sophia Children's Hospital.**

- Erasmus University Medical Center intends to contribute **€412.872,00** in cash towards for the first 2 years of the flagship for the project costs in accordance with the budget in the project proposal as follows:
    - 1 FTE postdoc 2 years Prinses Beatrix Spierfonds, 'Toward a clinical trial of lentiviral gene therapy for Pompe disease'
    - 2 FTE Brain Foundation + Metakids 'Hematopoietic stem cell-mediated lentiviral gene therapy for MPS II').

These contributions are from externally funded projects that align with the aims of the Flagship the Rare Disease Innovation Center

Yours sincerely,

Prof. Dr. Edmond H.H.M. Rings
Head of Department of Pediatrics
Erasmus MC Sophia Children's Hospital
Date: Friday March 11th, 2022

Postadres
Postbus 2060
3000 CB Rotterdam

Bezoekadres
Wytemaweg 80
3015 CN Rotterdam

Parkeergarage Wytema
Wytemaweg 12
3015 CN Rotterdam

Themadirecteur Thema Sophia
Mr. P. Beltman

Afdelingshoofd Klinische Genetica
Prof. dr. P.A.E. Sillevis Smitt

Afdelingshoofd Kindergeneeskunde
Prof.dr. E.H.H.M. Rings
Plv. Afdelingshoofd Kindergeneeskunde
Prof.dr. A.M.C. van Rossum

Afdelingshoofd Kinder- en Jeugdpsychiatrie/Psychologie wnmd. voorzitter
Prof.dr. M.H.J. Hillegers

Afdelingshoofd Kinderchirurgie/ Voorzitter Kinderchirurgische Groep
Prof.dr. R.M.H. Wijnen

Afdelingshoofd Urologie
Dr. J.L. Boormans

Themavoorzitter Thema Sophia/ Afdelingshoofd Verloskunde & Gynaecologie
Prof.dr. E.A.P. Steegers

Erasmus MC
Universitair Medisch Centrum Rotterdam

Prinses Beatrix Spierfonds, Brain Foundation and Metakids project leader in agreement with cash contribution to WP5:

Dr. W.W.M. Pim Pijnappel

Associate Professor Molecular Stem Cell Biology

Center for Lysosomal and Metabolic Diseases

Department of Pediatrics

Department of Clinical Genetics

Date: Wednesday March 9th, 2022

601787

Erasmus MC
University Medical Center Rotterdam

Pediatrics
Prof. E.H.H.M. Rings, MD, PhD
Prof. A.M.C. van Rossum, MD, PhD

Prof.dr. S. Sleijfer
voorzitter Raad van Bestuur Erasmus MC

Internal postal address **Sp 2471**
Date March 11, 2022

LETTER OF INTENT *for the* ***Rare Disease Innovation Center Flagship PROJECT***

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

Visiting address
Dr. Molewaterplein 40
3015 GD Rotterdam, NL

Contact & route
www.erasmusmc.nl

Dear Professor S. Sleijfer,

I, **Prof. Dr. Edmond Rings**, in my capacity of head of the department of Pediatrics at Erasmus University Medical Center, Erasmus MC Sophia Children's Hospital hereby confirm that *Erasmus University Medical Center* Rotterdam intends to contribute to ***the flagship the Rare Disease Innovation Center*** as applied for by the main applicant, **Dr. Marjon H. Cnossen, pediatric hematologist** at Erasmus MC Sophia Children's Hospital.

This funding comes from existing funding from the NWO-NWA SYMPHONY consortium which aims to orchestrate personalized treatment in patients with inherited rare bleeding disorders. Its aims nicely align with objectives of WPs in this Convergence Flagship call. Please see attachment/ NWO-NWA contract for details. The WPs 5 and 6 aim to implement value-added care by implementation of PROMs/ PROMIS and to personalize treatment by pharmacokinetic-guided dosing, respectively. WP8 will focus on costs of healthcare in patients with an inherited rare bleeding disorder. Therefore, these funds can serve as co-funding for the first two years of the consortium, as described in the grant application.

Erasmus University Medical Center intends to contribute in cash:

- From WP5: 0.5 fte PhD (2 years), is a total of: **€ 60.000,00 -> 2022: 30.000; 2023: 30.000**
- From WP6: 1 fte PhD (2 years), is a total of: **€ 120.000,00 -> 2022: 60.000; 2023: 60.000**
- From WP8: 0.5 fte PhD (2 years), is a total of: **€ 60.000,00 -> 2022: 30.000; 2023: 30.000**

Leading to a total contribution of **€ 240.000,00** which will function as co-financing for the first two years of the flagship.

Erasmus MC
Universitair Medisch Centrum Rotterdam

Yours sincerely,

Prof. Dr. Edmond H.H.M. Rings
Head of Department of Pediatrics

Erasmus MC Sophia Children's Hospital

Date: March 8, 2022

SYMPHONY workpackage leaders in agreement with alignment and in cash contribution.

Prof. Dr. Hester Lingsma

Head Medical Decision making

Erasmus MC

Date: 11-3-2022

Dr. Marjon H. Cnossen

Associate professor Pediatric Hematology

Erasmus MC Sophia Children's Hospital

Date: March 11, 2022

6017187

Date March 14, 2022
Contact Dr. Kateřina Staňková
Phone +31 (0)646 066 671
E-mail k.stankova@tudelft.nl
Subject Letter of Intent

Faculty of Technology, Policy and Management
Jaffalaan 5
2628 BX Delft

Prof. Dr. E.H.H.M. Rings
Head of Department of Pediatrics
Erasmus University Medical Center
Erasmus MC Sophia Children's Hospital
Wytemaweg 80
3015 GE Rotterdam
The Netherlands

Prof. Dr. E.H.H,M, Rings

I, **Dr. Kateřina Staňková**, in my capacity of a member of the Department of Engineering Systems and Services (ESS), Delft University of Technology, I hereby confirm that I intend to contribute to the flagship the Rare Disease Innovation Center, as applied for by the main applicant, **Dr. Marjon H. Cnossen, pediatric hematologist at Erasmus MC Sophia Children's Hospital**.

I intend to contribute **€93.714,00** in kind towards the project costs in accordance with the budget in the project proposal and budget from, leading to a total contribution of **€93.714,00** which will function as co-financing for the first two years of the flagship.

This contribution is from two projects that align with the flagship aims: European Training Network EvoGamesPlus, which focuses on application of strategic decision making also in medical domain and NWO KLEIN2 project, which focuses on improving cancer treatment by game theory and dynamical systems theory. Both these projects are led by me. The PhD student matched in kind is supervised by dr. Jafar Rezaei and me, the PostDoc matched is supervised by me.

Your sincerely,

Dr. Kateřina Staňková
Associate Professor in Dynamic Game Theory
Department of Engineering Systems and Services
Faculty of Technology, Policy and Management
Delft University of Technology

Biomedical Sciences
Dept. of Cell Biology

Prof. E.H.H.M. Rings
Dept. of Pediatrics
Erasmus MC

Direct dial +31 10 7043169
Fax number +31 10 7044743
Room number Ee1040
E-mail d.huylebroeck@erasmusmc.nl
Our reference NG/mvg
Date 7 March 2022

LETTER OF INTENT
for the
Rare Disease Innovation Center Flagship PROJECT

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

Street address
Wytemaweg 80
3015 CE Rotterdam, NL

Dear Prof. Rings,

I, Prof. Dr. Danny Huylebroeck, in my capacity of head of the Department of Cell Biology at Erasmus University Medical Center Hospital hereby confirm that *Erasmus University Medical Center* Rotterdam will contribute to the flagship the Rare Disease Innovation Center, as applied for by the main applicant, Dr. Marjon H. Cnossen, pediatric hematologist at Erasmus MC Sophia Children's Hospital.

Erasmus University Medical Center will contribute € 175,000 <u>in cash</u> towards the project costs in accordance with the budget in the project proposal and budget form. This represents 0.5 FTE OIO plus consumables.

This contribution is from *TRACER*, a joint project in the ZonMW stem cell programme PSIDER. The project was initiated by Erasmus MC and TU Delft and will run from 2022 until 2028. *TRACER* aims to develop technology for genetic correction of hematopoietic stem cells derived from patients with rare hereditary anemias, and develop these corrected hematopoietic stem cells as Advanced Therapy Medicinal Products for treatment of these patients. The *TRACER* project aligns seamlessly with the flagship aims.

Yours sincerely,

Prof. Dr. Danny Huylebroeck
Head of Department of Cell Biology

Erasmus MC
University Medical Center Rotterdam

Date: Monday March 7th, 2022

Other involved project leaders of *TRACER* agree with this in cash contribution.

Prof. Dr. Sjaak Philipsen, *TRACER* coordinator

Erasmus MC Department of Cell Biology

Date: Monday March 7th, 2022

Dr. Cees Haringa

Assistant professor Bioprocess Engineering

Faculty of Applied Sciences, TU Delft

Dr. Marieke Klijn

Assistant professor Bioprocess Engineering

Faculty of Applied Sciences, TU Delft

Erasmus MC
University Medical Center Rotterdam

Department of Clinical Genetics
Section Research and Education

Prof. Dr. S. Sleijfer
Chair Board of Directors Erasmus MC

Phone **0031** (0)10 7030051
E-mail t.barakat@erasmusmc.nl
Our reference 2022-43
Date 14/03/2022

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

Visiting address
Dr. Molewaterplein 40
3015 GD Rotterdam, NL

Contact & route
www.erasmusmc.nl

Concerning *LETTER OF INTENT Rare Disease Innovation Center Flagship*

Dear Dean,

I, Dr. Stefan Barakat, in my capacity of assistant professor of the department of Clinical Genetics at Erasmus University Medical Center, hereby confirm I intend to support and contribute to the flagship entitled “Rare Disease Innovation Center”, that we jointly apply for. Our Flagship will be led by Dr. Marjon H. Cnossen (main applicant, Erasmus MC), together with Armagan Albayrak (TU Delft) and myself as additional leads.

I intend to contribute towards the project costs in accordance with the budget attached to the project proposal:
-€130.000,- in cash (1 fte 2 years PhD).
-€111.861,60 in cash (0.3 fte 2 years PostDoc; 0,5 fte laboratory technician).
Together leading to a total contribution of €241.861,60 which will function as co-financing for the first two years of the Flagship. Both contributions are from externally funded projects (Stichting12q Research Fellowship; CURE Epilepsy Award / EpilepsieNL Grant) that align with the aims of the flagship the Rare Disease Innovation Center.

Sincerely,

T. S. Barakat

Dr. T.S. Barakat PhD, MD, MSc
Assistant professor
Head of the non-coding genome research group
Clinical Genetics

https://www.erasmusmc.nl/en/research/groups/barakat-lab-non-coding-genome-in-clinical-genetics

VSOP

PATIËNTENKOEPEL VOOR ZELDZAME EN GENETISCHE AANDOENINGEN

Soest, 11 maart 2022

Prof. Dr. E.H.H.M. Rings
Head of Department of Pediatrics
Erasmus University Medical Center, Erasmus MC Sophia Children's Hospital
Wytemaweg 80
3015 GE Rotterdam
The Netherlands

Letter of Intent for the Rare Disease Innovation Center Flagship PROJECT

Dear Prof Rings,

I, Dr. Cor Oosterwijk, in my capacity of director of VSOP – Patiëntenorganisatie voor zeldzame en genetische aandoeningen, hereby confirm that VSOP will to contribute to the flagship the Rare Disease Innovation Center, as applied for by the main applicant, Dr. Marjon H. Cnossen, pediatric hematologist at Erasmus MC Sophia Children's Hospital.

Yours sincerely,

Dr. Cor Oosterwijk
Director

PHARMO

Van Deventerlaan 30-40
3528 AE Utrecht
The Netherlands

+31 (0)30 744 08 00
info@pharmo.com
www.pharmo.com

Erasmus MC
Marjon Cnossen

Date
10-03-2022

Contactperson
Ella Jansen

Subject
Letter of intent for the Flagship Project

Telefoonnummer
+31 30-7440800

Dear Marjon Cnossen,

I, Ron Herings, in my capacity of Scientific Director at the PHARMO Institute hereby confirm that the PHARMO Institute intends to contribute to the Flagship project, as applied for by the main applicant, Marjon Cnossen, Pediatric Hematologist and Associate Professor Pediatric Hematology at Erasmus MC.

The PHARMO Institute intends to contribute € 0,- in cash towards the project costs in accordance with the budget in the project proposal and budget form.

The PHARMO Institute and STIZON intends to provide an in-kind contribution of access to their data infrastructure, representing a monetary value of € 50.000,- and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Ron Herings

Position: Scientific Director

Date: 10-03-2022

Genome scan

Plesmanlaan 1d
2333 BZ Leiden
The Netherlands

Telephone +31 71 568 1050
Fax +31 71 568 1055

E-mail info@genomescan.nl
Web www.genomescan.nl

To: Erasmus Medical Center
Prof.dr. S. Sleijfer
Dean
Postbus 2040
3000 CA Rotterdam

Subject: **LETTER OF INTENT for the Rare Disease Innovation Center Flagship PROJECT**
Date: 14-03-2022

Dear Prof. Sleijfer,

I, Kees van den Berg, in my capacity of CEO of GenomeScan BV., hereby confirm that GenomeScan BV intends to support the **Rare Disease Innovation Center** Flagship (**RDIC**) project, as applied for by the main applicant, Stefan Barakat, assistant professor at the Department of Clinical Genetics, Erasmus MC.

It is a widely appreciated that patients suffering from a rare disease unfortunately are falling through the cracks relatively frequent. Returning issues are that of increased risk, ever rising costs and lack of novel therapies. To tackle such limitations significantly hampering quality of life, a proper diagnostic setup is crucial. In this regard, identification of genomic, epigenomic and transcriptomic biomarkers has over the last decade proven to be of profound use in stratifying patient groups, as it paves the way for improved patient care and intervention.

GenomeScan is devoted to improve healthcare by providing NGS services and bioinformatics solutions under the scope of ISO/IEC 17025 [L518] and ISO15189 [M308] accreditation. We continuously seek collaboration with both clinicians and researchers to remain at the forefront of scientific breakthrough, and to create opportunities for innovation that allows for strengthening our expertise in NGS technologies. Utilizing our NGS services and knowledge, we intend and foresee to play an important advisory role in the context of this initiative.

GenomeScan will support this project with its expert knowledge of NGS related to Rare Disease Diagnostics and Innovation and will take place in the advisory board.

At GenomeScan we believe that improving the journey of patients suffering from a rare disease needs prioritization, as it will affect not only their quality of life but also increase cost effectiveness, and in general could serve as a footprint for understanding other disease-related problems. I therefore fully support the RDIC project effort.

Yours sincerely,

Kees van den Berg
CEO

Annex III - Letter of support from department head of the main lead

Erasmus MC
University Medical Center Rotterdam

Pediatrics
Prof. E.H.H.M. Rings, MD, PhD
Prof. A.M.C. van Rossum, MD, PhD

Prof.dr. S. Sleijfer
voorzitter Raad van Bestuur Erasmus MC

Internal postal address Sp 2471
Date March 9, 2022

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

Visiting address
Dr. Molewaterplein 40
3015 GD Rotterdam, NL

Contact & route
www.erasmusmc.nl

LETTER OF INTENT** for the **Rare Disease Innovation Center Flagship PROJECT

Dear Professor S. Sleijfer,

I, **Prof. Dr. Edmond Rings**, in my capacity of head of the department of Pediatrics at Erasmus University Medical Center, Erasmus MC Sophia Children's Hospital hereby confirm that *Erasmus University Medical Center* Rotterdam supports ***the Flagship the Rare Disease Innovation Center*** as applied for by the main applicant, **Dr. Marjon H. Cnossen, pediatric hematologist** at Erasmus MC Sophia Children's Hospital.

Yours sincerely,

Prof. Dr. Edmond H.H.M. Rings
Head of Department of Pediatrics

Erasmus MC Sophia Children's Hospital

Date: Tuesday March 8th, 2022

Erasmus MC
University Medical Center Rotterdam

Pediatrics
Prof. E.H.H.M. Rings, MD, PhD
Prof. A.M.C. van Rossum, MD, PhD

Prof.dr. S. Sleijfer
voorzitter Raad van Bestuur Erasmus MC

Internal postal address Sp 2471
Date March 9, 2022

LETTER OF INTENT *for the* ***Rare Disease Innovation Center Flagship PROJECT***

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

Visiting address
Dr. Molewaterplein 40
3015 GD Rotterdam, NL

Contact & route
www.erasmusmc.nl

Dear Professor S. Sleijfer,

I, **Prof. Dr. Edmond Rings**, in my capacity of head of the department of Pediatrics at Erasmus University Medical Center, Erasmus MC Sophia Children's Hospital, hereby confirm that *Erasmus University Medical Center* Rotterdam supports ***the Flagship the Rare Disease Innovation Center*** as applied for by the main applicant, **Dr. Marjon H. Cnossen, pediatric hematologist** at Erasmus MC Sophia Children's Hospital.

As this Flagship is a merge of two prior Flagships, the head of Department of Clinical Genetics also states his support after this merge by signing this letter of intent (Lead of prior Clinical Genetic Flagship: Dr. Stefan Barakat).

Yours sincerely,

Edmond Rings

Prof. Dr. Edmond H.H.M. Rings
Head of Department of Pediatrics

Erasmus MC Sophia Children's Hospital

Date: Wednesday March 9th, 2022

Prof. Dr. P.A.E. Sillevis Smitt

Head of Department of Clinical Genetics

Erasmus MC

Date: Wednesday March 9th, 2022